PROBABILITY III

Lakshmi Priya M E, notes by Ramdas Singh

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Chapter 1

REFINING THE NOTION OF PROBABILITY

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Consider the random walk on the line; a drunk ant sits at the origin and takes steps of length 1 to the left or right with equal probability. We wish to obtain the probability that after an even n steps, the ant returns to the origin. Of course, the natural approach is to frame this problem in a discrete probability space.

Definition 1.1. A *discrete probability space* is a pair (Ω, q) where Ω is a non-empty finite or countably infinite set called the sample space, and $q : \Omega \to [0,1]$ is a function satisfying $\sum_{\omega\in\Omega} q(\omega) = 1$. Any subset of Ω is termed an event, and the probability of an event $A \subseteq \Omega$ is defined as $\mathbb{P}(A) = \sum_{\omega\in A} q(\omega)$.

For the problem above, we can take the sample space to be

$$\Omega = \{\mathbf{x} = (x_0 = 0, x_1, x_2, \ldots, x_n) \in \mathbb{Z}^{n+1} \mid x_{i+1} - x_i \in \{-1, 1\} \text{ for all } 0 \le i < n\}, \tag{1.1}$$

and the mapping q to be $q(\mathbf{x}) = 2^{-n}$ for all $\mathbf{x} \in \Omega$. Then, the event that the ant returns to the origin after n steps is given by $A = \{\mathbf{x} \in \Omega \mid x_n = 0\}$, and we have

$$\mathbb{P}(A) = \sum_{\mathbf{x}\in A} q(\mathbf{x}) = 2^{-n}\,|A| = 2^{-n}\binom{n}{n/2}. \tag{1.2}$$

Reframing the problem, let us consider a random walk on the line starting at the origin, and continuing indefinitely. In this case, the sample space is given by

$$\Omega = \{\mathbf{x} = (x_0 = 0, x_1, x_2, \ldots) \in \mathbb{Z}^{\mathbb{N}} \mid x_{i+1} - x_i \in \{-1, 1\} \text{ for all } i \in \mathbb{N}\}. \tag{1.3}$$

This Ω is uncountably infinite. In such a case, we cannot define a probability measure by assigning a positive probability to each element of Ω. As another example of difficulty in dealing with uncountable sample spaces, consider the problem of choosing a real number uniformly at random from the interval $[0,1]$. If we attempt to assign a positive probability to each element of $[0,1]$, we would have to assign a probability of 0 to each element, since there are uncountably many elements in $[0,1]$. However, this would lead to a total probability of 0, which is not valid. Of course, we know that the probability of choosing a specific element or a countable set of elements from $[0,1]$ is 0, but there then arises the question of how to assign a probability to an uncountable set of elements, such as the Cantor set. We can work in a slightly different viewpoint.

Definition 1.2. A probability space is a tuple $(\Omega, \mathcal{F}, \mathbb{P})$ where Ω is a non-empty set called the sample space, $\mathcal{F}$ is the collection of all subsets of Ω ($\sim 2^{\Omega}$), and $\mathbb{P} : \mathcal{F} \to [0,1]$ is a function satisfying $\mathbb{P}(\Omega) = 1$ and countable additivity, that is, for any countable collection of pairwise disjoint sets $A_1, A_2, \ldots \in \mathcal{F}$,

we have

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i). \tag{1.4}$$

Using this definition, we look to define a $\mathbb{P}$ on subsets of $[0,1]$ that satisfies the above properties, along with the property that $\mathbb{P}([a,b]) = b - a$. Thus, we have $\mathbb{P}([0,1]) = 1$. If we have two disjoint intervals $I_1, I_2 \subseteq [0,1]$, then $\mathbb{P}(I_1 \cup I_2) = \mathbb{P}(I_1) + \mathbb{P}(I_2)$. This is a natural property to expect from a probability measure. What about a general set $S \subseteq [0,1]$? One can observe that if we have a sequence of intervals $I_1, I_2, \ldots$ such that $S \subseteq \bigcup_{k=1}^{\infty} I_k$, then we would have $\mathbb{P}(S) \leq \sum_{k=1}^{\infty} \mathbb{P}(I_k)$. This is because the probability of S should be less than or equal to the probability of any set that contains S. Thus, we can define

$$\mathbb{P}_*(S) = \inf\left\{\sum_{k=1}^{\infty} |I_k| \;\middle|\; S \subseteq \bigcup_{k=1}^{\infty} I_k,\ I_k \subseteq [0,1]\right\}. \tag{1.5}$$

We have not verified that this is a valid definition of a probability measure, hence the subscript $*$. Via this, let us try computing $\mathbb{P}_*(\mathbb{Q} \cap [0,1])$. Enumerate the rationals in $[0,1]$ as $\{q_1, q_2, \ldots\}$. For any $\epsilon > 0$, we can choose intervals $I_k = [q_k - \epsilon/2^{k+1}, q_k + \epsilon/2^{k+1}] \cap [0,1]$ for each $k \in \mathbb{N}$. Note that $\bigcup_{k=1}^{\infty} I_k$ contains all the rationals in $[0,1]$, and $\sum_{k=1} |I_k| \leq \sum_{k=1}^{\infty} \epsilon/2^k = \epsilon$. Since ϵ was arbitrary, we have $\mathbb{P}_*(\mathbb{Q} \cap [0,1]) = 0$. This is consistent with our intuition that the probability of choosing a rational number from $[0,1]$ is 0. So, $\mathbb{P}_*$ seems to be a reasonable definition of a probability measure on $([0,1], 2^{[0,1]})$. However, we will show later that there exists a set $\mathcal{A} \subseteq [0,1]$ such that $\mathbb{P}_*(\mathcal{A}) = \mathbb{P}_*([0,1] \setminus \mathcal{A}) = 1$, contradicting the property that $\mathbb{P}_*([0,1]) = 1$. We state the following theorem without proof.

Theorem 1.3. *There is no probability measure $\mathbb{P}$ on $([0,1], 2^{[0,1]})$ such that $\mathbb{P}([a,b]) = b - a$ for all $[a,b] \subseteq [0,1]$.*

If we remove the requirement that the probability measure of a subinterval $[a,b]$ is equal to its length, then we can define a probability measure on $([0,1], 2^{[0,1]})$. For example, we can define $\mathbb{P}(S) = 1$ if $1/2 \in S$ and $\mathbb{P}(S) = 0$ otherwise. This is a valid probability measure, but it is not very useful. Keeping the requirement that the probability measure of a subinterval $[a,b]$ is equal to its length, we can instead restrict the collection of subsets of $[0,1]$ on which we define the probability measure.

Definition 1.4. A *probability space* is a tuple $(\Omega, \mathcal{F}, \mathbb{P})$ where

- Ω is a non-empty set called the sample space,
- $\mathcal{F}$ is a collection of subsets of Ω satisfying the following properties:
  - $\Omega \in \mathcal{F}$,
  - if $A \in \mathcal{F}$, then $\Omega \setminus A \in \mathcal{F}$,
  - if $A_1, A_2, \ldots \in \mathcal{F}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$,
- $\mathbb{P} : \mathcal{F} \to [0,1]$ is a function satisfying:
  - $\mathbb{P}(\Omega) = 1$,
  - if $A_1, A_2, \ldots \in \mathcal{F}$ are pairwise disjoint, then $\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$.

Such an $\mathcal{F}$ is termd a *σ-algebra* on Ω, and such a $\mathbb{P}$ is termed a *probability measure* on $(\Omega, \mathcal{F})$.

1.1 σ-algebras

Example 1.5. Given Ω, trivial possible choices for $\mathcal{F}$ are $\{\emptyset, \Omega\}$ and 2^{Ω}. A more interesting example is the following: suppose Ω is uncountable. Then $\mathcal{F} = \{A \subseteq \Omega \mid A \text{ is countable or } \Omega \setminus A \text{ is countable}\}$ is a σ-algebra on Ω.

Let $S \subseteq 2^\Omega$ be a collection of subsets of Ω. We claim that there exists a notion of the smallest σ-algebra containing S. This is called the σ-algebra generated by S, and is denoted by $\sigma(S)$.

Lemma 1.6. *For any collection $S \subseteq 2^\Omega$, there exists a unique smallest σ-algebra containing S satisfying*

$$\sigma(S) = \bigcap \{\mathcal{F} \subseteq 2^\Omega \mid \mathcal{F} \text{ is a } \sigma\text{-algebra and } S \subseteq \mathcal{F}\}. \tag{1.6}$$

Proof. It is easy to check that $\sigma(S)$ is indeed a σ-algebra containing S. If $\mathcal{F}$ is any σ-algebra containing S, then $\sigma(S) \subseteq \mathcal{F}$. Thus, $\sigma(S)$ is the unique smallest σ-algebra containing S. ∎

Definition 1.7. Let (X, d) be a metric space (or a topological space), and let $\mathcal{S} \subseteq 2^X$ be the colletion of all open sets in X. Then, the σ-algebra $\sigma(\mathcal{S})$ is called the *Borel σ-algebra* on X, and its elements are called *Borel sets*.

We will show that there exists a probability measure on $([0,1], \mathcal{B})$, where $\mathcal{B}$ is the Borel σ-algebra on $[0,1]$, which extends the notion of length of intervals to Borel sets.

Let us return to the problem of indefinite random walks on the line. Let $\Omega = \{\omega = (\omega_1, \omega_2, \ldots) \mid \omega_i \in \{0,1\}\}$, where $\omega_i = 0$ corresponds to a step to the left and $\omega_i = 1$ corresponds to a step to the right. If we let $A = \{0,1\}$, then $\Omega = A^{\mathbb{N}}$; consider the discrete topology on A, and the product topology on Ω. If we let

$$\mathcal{S} = \{U_1 \times U_2 \times \cdots \times U_k \times A \times A \cdots \mid U_i = \{0\} \text{ or } \{1\}\} \tag{1.7}$$

then one can show that the σ-algebra generated by $\mathcal{S}$ is the Borel σ-algebra on Ω. We can set $\mathcal{F} = \sigma(\mathcal{S})$, and define a probability measure $\mathbb{P}$ on $(\Omega, \mathcal{F})$ by setting $\mathbb{P}(U_1 \times U_2 \times \cdots \times U_k \times A \times A \cdots) = 2^{-k}$ for all $U_i = \{0\}$ or $\{1\}$.

July 24th.

Example 1.8. Let (X, d) be a separable metric space (that is, there exists a countable dense subset). The Borel σ-algebra on X is then

$$\mathcal{B}(X) = \sigma(\{B(p,r)\}_{p \in X, r>0}). \tag{1.8}$$

Note that the usual containment is $\sigma(\{B(p,r)\}_{p \in X, r>0}) \subseteq \mathcal{B}(X)$ for a metric space X. So, it suffices to show that if $V \in \tau$, the topology on X, then $V \in \sigma(\{B(p,r)\}_{p \in X, r>0})$. Let Q be a countable dense subset of X. Then, for any $V \in \tau$, we have

$$V = \bigcup_{n=1}^{\infty} B(q_n, r_n), \tag{1.9}$$

where $q_n \in Q \cap V$ and $r_n > 0$ (where r_n is the largest possible radius such that $B(q_n, r_n) \subseteq V$). Since $\sigma(\{B(p,r)\}_{p \in X, r>0})$ is a σ-algebra, we have $V \in \sigma(\{B(p,r)\}_{p \in X, r>0})$. Thus, we have $\mathcal{B}(X) = \sigma(\{B(p,r)\}_{p \in X, r>0})$.

Example 1.9. Let $(\Sigma, \mathcal{F}, \mathbb{P})$ be a probability space.

- $\mathcal{F}$ is closed under countable unions, countable intersections, differences, and symmetric differences. Countable unions and intersections follow from the definition of a σ-algebra. If $A, B \in \mathcal{F}$, then $A \setminus B = A \cap (\Omega \setminus B) \in \mathcal{F}$. Finally, $A \triangle B = (A \setminus B) \cup (B \setminus A) \in \mathcal{F}$.
- Let $(\Sigma, \mathcal{F}, \mathbb{P})$ be a probability space, and let $A_1, A_2, \ldots \in \mathcal{F}$. Define

$$\limsup_{n \to \infty} A_n := \{x \in \Omega \mid x \text{ belongs to infinitely many } A_n\} \tag{1.10}$$

and

$$\liminf_{n \to \infty} A_n := \{x \in \Omega \mid x \text{ belongs to all but finitely many } A_n\}. \tag{1.11}$$

Then, both the lim sup and lim inf belong in $\mathcal{F}$. To see this, simply note that

$$\limsup_{n\to\infty} A_n = \bigcap_{n=1}^{\infty}\bigcup_{k=n}^{\infty} A_k, \quad \liminf_{n\to\infty} A_n = \bigcup_{n=1}^{\infty}\bigcap_{k=n}^{\infty} A_k. \tag{1.12}$$

- One may show that $\mathbb{P}(\emptyset) = 0$, $\mathbb{P}(A) \leq \mathbb{P}(B)$ if $A \subseteq B$, $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$, and $\mathbb{P}(\bigcup A_n) \leq \sum \mathbb{P}(A_n)$ for any countable collection of sets $A_1, A_2, \ldots \in \mathcal{F}$.
- For a nested sequence of sets $A_1 \subseteq A_2 \subseteq \cdots$ in $\mathcal{F}$, let $A = \bigcup_{n=1}^{\infty} A_n$ (this is usually denoted by $A_n \uparrow A$). Then, $\mathbb{P}(A) = \lim_{n\to\infty} \mathbb{P}(A_n)$. To this end, let $B_n = A_n \setminus A_{n-1}$ for $n > 1$, and $B_1 = A_1$. Then $A = \bigcup_{n=1}^{\infty} B_n$. Thus,

$$\mathbb{P}(A) = \sum_{n=1}^{\infty} \mathbb{P}(B_n) = \lim_{N\to\infty} \sum_{n=1}^{N} \mathbb{P}(B_n) = \lim_{N\to\infty} \sum_{n=2}^{N} \mathbb{P}(A_n) - \mathbb{P}(A_{n-1}) + \mathbb{P}(A_1) = \lim_{N\to\infty} \mathbb{P}(A_N). \tag{1.13}$$

1.1.1 Other Systems of Sets

The following are other interesting collection of subsets of Ω besides the Borel σ-algebra.

1. *π-system*: a collection of subsets $S \subseteq \Omega$ such that $A, B \in S$ implies $A \cap B \in S$.
2. *λ-system*: a collection Λ of subsets of Ω such that $\Omega \in \Lambda$, if $A, B \in \Lambda$ with $A \subseteq B$, then $B \setminus A \in \Lambda$, and if $A_1, A_2, \ldots \in \Lambda$, then $\bigcup_{n=1}^{\infty} A_n \in \Lambda$.
3. *algebra*: a collection $\mathcal{A}$ of subsets of Ω such that $\Omega \in \mathcal{A}$, if $A \in \mathcal{A}$, then $\Omega \setminus A \in \mathcal{A}$, and if $A, B \in \mathcal{A}$, then $A \cup B \in \mathcal{A}$.

We had defined $\omega(S)$, the σ-algebra generated by a collection of subsets $S \subseteq 2^{\Omega}$. We can similarly define $\pi(S)$, the π-system generated by S, $\lambda(S)$, the λ-system generated by S, and $\mathcal{A}(S)$, the algebra generated by S.

Theorem 1.10 (*Sierpiński-Dynkin π-λ Theorem*)**.** *Let $S, \mathcal{F} \subseteq 2^{\Omega}$ be collections of subsets of Ω. Then,*

1. *$\mathcal{F}$ is a σ-algebra if and only if $\mathcal{F}$ is both a π-system and a λ-system.*
2. *If S is a π-system, then $\lambda(S) = \sigma(S)$.*

Proof. 1. If $\mathcal{F}$ is a σ-algebra, then it is trivially both a π-system and a λ-system. Conversely, if $\mathcal{F}$ is both a π-system and a λ-system, we need to verify the axioms of a σ-algebra. We have $\Omega \in \mathcal{F}$ since $\mathcal{F}$ is a λ-system. If $A \in \mathcal{F}$, then $\Omega \setminus A \in \mathcal{F}$ since $\mathcal{F}$ is a λ-system. Finally, if $A_1, A_2, \ldots \in \mathcal{F}$, then we can write $B_n = \bigcup_{i=1}^{n} A_i \in \mathcal{F}$ since $A_i^c \in \mathcal{F}$ and $B_n^c = \bigcap_{i=1}^{n} A_i^c \in \mathcal{F}$ since $\mathcal{F}$ is a π-system. Then, $B_n \uparrow B = \bigcup_{i=1}^{\infty} A_i$, and since $\mathcal{F}$ is a λ-system, we have $B \in \mathcal{F}$. Thus, $\mathcal{F}$ is a σ-algebra.

2. Let S be a π-system. Since $\lambda(S) \subseteq \sigma(S)$, it suffices to show that $\lambda(S)$ is a π-system. Fix $A \in S$, and define $\mathcal{C}_A = \{B \in \lambda(S) \mid A \cap B \in \lambda(S)\}$. Note that $S \subseteq \mathcal{C}_A$ because S is a π-system. We show that $\mathcal{C}_A$ is a λ-system. We have $\Omega \in \mathcal{C}_A$ since $A \cap \Omega = A \in \lambda(S)$. If $B_1, B_2 \in \mathcal{C}_A$ with $B_1 \subseteq B_2$, then $A \cap B_1 \subseteq A \cap B_2$, and since $A \cap B_2 \in \lambda(S)$, we have $(A \cap B_2) \setminus (A \cap B_1) = A \cap (B_2 \setminus B_1) \in \lambda(S)$, and thus $B_2 \setminus B_1 \in \mathcal{C}_A$. Finally, if $B_1, B_2, \ldots$ are in $\mathcal{C}_A$, then $A \cap B_n \in \lambda(S)$ for all n, and thus $\bigcup(A \cap B_n) = A \cap (\bigcup B_n) \in \lambda(S)$, and hence $\bigcup B_n \in \mathcal{C}_A$. Thus, $\mathcal{C}_A$ is a λ-system. Since $\lambda(S)$ is the smallest λ-system containing S, we have $\lambda(S) = \mathcal{C}_A$, and hence for any $B, C \in S$, we have $B, C \in S$ implies $B \cap C = A \cap C \in S$. Thus, $\lambda(S)$ is a π-system.

■

Lemma 1.11. *Let S be a π-system, and let $\mathbb{P}$ and $\mathbb{Q}$ be two probability measures on $(\Sigma, \sigma(S))$ such that $\mathbb{P}(A) = \mathbb{Q}(A)$ for all $A \in S$. Then, $\mathbb{P}(A) = \mathbb{Q}(A)$ for all $A \in \sigma(S)$.*

Proof. Let $\mathcal{C} = \{B \in \sigma(S) \mid \mathbb{P}(B) = \mathbb{Q}(B)\}$. Then, $S \subseteq \mathcal{C}$, and we show that $\mathcal{C}$ is a λ-system. We have $\Omega \in \mathcal{C}$ since $\mathbb{P}(\Omega) = 1 = \mathbb{Q}(\Omega)$. If $B_1, B_2 \in \mathcal{C}$ with $B_1 \subseteq B_2$, then $\mathbb{P}(B_2 \setminus B_1) = \mathbb{P}(B_2) - \mathbb{P}(B_1) = \mathbb{Q}(B_2) - \mathbb{Q}(B_1) = \mathbb{Q}(B_2 \setminus B_1)$, and hence $B_2 \setminus B_1 \in \mathcal{C}$. Finally, if $B_1, B_2, \ldots$ are in $\mathcal{C}$, then $\mathbb{P}(A) = \lim_{n\to\infty} \mathbb{P}(A_n)$, and since $\mathbb{Q}(A) = \lim_{n\to\infty} \mathbb{Q}(A_n)$, we have $\mathbb{P}(A) = \mathbb{Q}(A)$. ■

July 28th.

Example 1.12. We wish to obtain an example of S which is *not* a π-system, and two probability measures $\mathbb{P}$ and $\mathbb{Q}$ on $(\Sigma, \sigma(S))$ such that $\mathbb{P}(A) = \mathbb{Q}(A)$ for all $A \in S$, but $\mathbb{P}(A) \neq \mathbb{Q}(A)$ for some $A \in \sigma(S)$. Let $\Omega = \{1, 2, 3, 4\}$, and let $S = \{\{1, 2\}, \{2, 3\}\}$. Then, $\sigma(S) = 2^\Omega$. Define $\mathbb{P}$ and $\mathbb{Q}$ on $(\Omega, \sigma(S))$ by

$$\mathbb{P}(\{1\}) = \mathbb{P}(\{2\}) = \mathbb{P}(\{3\}) = \mathbb{P}(\{4\}) = 1/4, \tag{1.14}$$

and

$$\mathbb{Q}(\{1\}) = \mathbb{Q}(\{3\}) = 1/6, \quad \mathbb{Q}(\{2\}) = \mathbb{Q}(\{4\}) = 1/3. \tag{1.15}$$

Then, $\mathbb{P}(\{1, 2\}) = \mathbb{Q}(\{1, 2\}) = 1/2$ and $\mathbb{P}(\{2, 3\}) = \mathbb{Q}(\{2, 3\}) = 1/2$, but $\mathbb{P}(\{3\}) = 1/4 \neq 1/6 = \mathbb{Q}(\{3\})$.

Example 1.13. Let $\Omega \neq \emptyset$, and A_λ be a collection of subsets of Ω indexed by $\lambda \in \Lambda$. Consider the σ-algebra $\sigma(\{A_\lambda\}_{\lambda\in\Lambda})$, and pick $B \in \sigma(\{A_\lambda\}_{\lambda\in\Lambda})$. We will show that there exists a countable subcollection $\{A_{\lambda_n}\}_{n=1}^\infty$ such that $B \in \sigma(\{A_{\lambda_n}\}_{n=1}^\infty)$. Let $\mathcal{C} = \{B \in \sigma(\{A_\lambda\}_{\lambda\in\Lambda}) \mid B \in \sigma(\{A_{\lambda_n}\}_{n=1}^\infty)$ for some countable subcollection $\{A_{\lambda_n}\}_{n=1}^\infty\}$. Then, $\mathcal{C}$ is a σ-algebra containing $\{A_\lambda\}_{\lambda\in\Lambda}$, and hence $\sigma(\{A_\lambda\}_{\lambda\in\Lambda}) \subseteq \mathcal{C}$. Thus, $B \in \mathcal{C}$, and hence there exists a countable subcollection $\{A_{\lambda_n}\}_{n=1}^\infty$ such that $B \in \sigma(\{A_{\lambda_n}\}_{n=1}^\infty)$.

1.2 The Lebesgue Measure

Theorem 1.14. *There exists a unique Borel measure λ on $[0, 1]$ such that $\lambda(I) = |I|$ for all intervals $I \subseteq [0, 1]$.*

Proof. For the π-system $S = \{(a, b] \mid 0 \leq a < b \leq 1\}$, we know that $\sigma(S)$ is a Boreal σ-algebra on $[0, 1]$. If λ and μ are two Borel measures on $[0, 1]$ such that $\lambda(I) = \mu(I) = |I|$ for all intervals $I \subseteq [0, 1]$, then by the previous lemma, we have $\lambda(A) = \mu(A)$ for all $A \in \sigma(S)$. Thus, the Borel measure is unique.

To show the existence of such a measure, we will define a pseudomeasure λ_* that satisfies the interval property, and then show that it is a measure on the Borel σ-algebra. For any $E \subseteq [0, 1]$, define

$$\lambda_*(E) = \inf\left\{\sum_{i=1}^\infty |I_i| \;\middle|\; I_i \text{ are intervals and } E \subseteq \bigcup_{i=1}^\infty I_i\right\} \tag{1.16}$$

Infer that $0 \leq \lambda_*(E) \leq 1$ for all $E \subseteq [0, 1]$. Moreover, let A and B be two sets in $[0, 1]$. Then, for any $\varepsilon > 0$, we have

$$\lambda_*(A \cup B) \leq \sum_{k=1}^\infty |I_k| + \sum_{k=1}^\infty |J_k| \leq \lambda_*(A) + \varepsilon + \lambda_*(B) + \varepsilon \implies \lambda_*(A \cup B) \leq \lambda_*(A) + \lambda_*(B). \tag{1.17}$$

Finally, let $(a, b] \subseteq [0, 1]$ be an interval. Then, $\lambda_*((a, b]) = b - a$. To see this, suppose $(a, b] \subseteq \bigcup_{k=1}^\infty I_k$ for some intervals I_k. Now consider $[a + \varepsilon, b]$ for some $\varepsilon > 0$ such that $[a + \varepsilon, b] \subseteq \bigcup_{k=1}^\infty I_k$. This is a compact subinterval, so we can extract a finite subcover $[a + \varepsilon, b] \subseteq \bigcup_{k=1}^N I_k$. Then,

$$b - a - \varepsilon = |[a + \varepsilon, b]| \leq \sum_{k=1}^N |I_k| \leq \sum_{k=1}^\infty |I_k| \implies b - a \leq \lambda_*((a, b]). \tag{1.18}$$

Now, since $(a,b] \subseteq (a, b+\varepsilon)$, a similar line of reasoning shows that $\lambda_*((a,b]) \leq b-a$. Thus, we have $\lambda_*((a,b]) = b-a$. The property of finite subadditivity also implies the property of countable subadditivity (show this). We now construct the σ-algebra $\mathcal{F}$ on which λ_* is a measure. We pick

$$\mathcal{F} = \{A \subseteq [0,1] \mid \lambda_*(E) = \lambda_*(E \cap A) + \lambda_*(E \cap A^c) \text{ for all } E \subseteq [0,1]\}. \tag{1.19}$$

We show that $\mathcal{F}$ is a σ-algebra containing the Borel σ-algebra, and that λ_* is a measure on $\mathcal{F}$. It is left as a minor exercise to show that $\mathcal{F}$ is closed under intersections and complements (making $\mathcal{F}$ a π-system). We show $\mathcal{F}$ is a λ-system. Let $A, B \in \mathcal{F}$. Then $B \setminus A = B \cap A^c$ is in $\mathcal{F}$. Then, for any $E \subseteq [0,1]$, we have

$$\lambda_*(E) = \lambda_*(E \cap A_n) + \lambda_*(E \cap A_n^c) \tag{1.20}$$
$$\geq \lambda_*(E \cap A_n) + \lambda_*(E \cap A^c), \tag{1.21}$$

because $A^c \subseteq A_n^c$. Since $A_n \uparrow A$, we have $E \cap A_n \uparrow E \cap A$, and hence

$$\lambda_*(E \cap A_n) \uparrow \lambda_*(E \cap A).$$

Passing to the limit gives

$$\lambda_*(E) \geq \lambda_*(E \cap A) + \lambda_*(E \cap A^c).$$

On the other hand, subadditivity of the outer measure gives

$$\lambda_*(E) \leq \lambda_*(E \cap A) + \lambda_*(E \cap A^c).$$

Therefore,

$$\lambda_*(E) = \lambda_*(E \cap A) + \lambda_*(E \cap A^c),$$

so $A \in \mathcal{F}$. ■

July 30th.

Note that $\mathcal{F} \subsetneq 2^{[0,1]}$; that is, there exist subsets of $[0,1]$ that are not in $\mathcal{F}$. We shall assume that $\lambda_*(E) = \lambda_*(E + x)$ for all $E \subseteq [0,1]$ and $x \in \mathbb{R}$ such that $E + x \subseteq [0,1]$. This is a reasonable assumption, since we expect the Lebesgue measure to be translation invariant (one can prove this). Moreover, instead of looking at $[0,1]$ we shall look at $([0,1), \oplus)$, where $\oplus$ is addition modulo 1. Now suppose we manage to break $[0,1)$ into countably many disjoint pieces $A_1, A_2, \ldots$ such that $\bigcup_{n=1}^{\infty} A_n = [0,1)$, and $A_i = A_1 \oplus r_i$ for some $r_i \in [0,1)$. Then, we would have $\lambda_*(A_i) = \lambda_*(A_1)$ for all i, and hence $\lambda_*([0,1)) = \sum_{n=1}^{\infty} \lambda_*(A_n) = \sum_{n=1}^{\infty} \lambda_*(A_1)$. This is a contradiction, since $\lambda_*([0,1)) = 1$ and $\sum_{n=1}^{\infty} \lambda_*(A_1)$ is either 0 or ∞. Thus, we have shown that there exists a set $A \subseteq [0,1)$ such that $A \notin \mathcal{F}$.

To construct these A_i's, simply look at the equivalence relation $\sim$ on $[0,1)$ defined by $x \sim y$ if and only if $x - y \in \mathbb{Q}$. Then, we can pick a representative from each equivalence class to form a set A_1. Then, we can define $A_i = A_1 \oplus r_i$ for some $r_i \in [0,1)$ such that $\{r_i\}_{i=1}^{\infty}$ is a complete set of representatives of $\mathbb{Q} \cap [0,1)$. This gives us the desired decomposition of $[0,1)$ into disjoint pieces.

1.2.1 Carathéodory's Extension Theorem

August 4th.

Suppose Ω is non-empty, and $\mathcal{A}$ is an *algebra*; that is, $\mathcal{A}$ is a collection of subsets of Ω such that $\Omega \in \mathcal{A}$, and $\mathcal{A}$ is closed under complements and finite unions. Further say that μ is a *pre-measure* on $\mathcal{A}$; that is, $\mu : \mathcal{A} \to [0,1]$ satisfies $\mu(\emptyset) = 0$, and if $A_1, A_2, \ldots \in \mathcal{A}$ are pairwise disjoint, then $\mu(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mu(A_i)$. Then, we can extend μ to a measure on the σ-algebra generated by $\mathcal{A}$. This is the content of *Carathéodory's extension theorem*.

Remark 1.15.
1. Note that since $\mathcal{A}$ is a π-system, such an extension must be unique.
2. For a π-system S, a pre-measure on § gives a pre-measure on $\mathcal{A}(S)$, the algebra generated by S, since every set $B \in \mathcal{A}(S)$ can be written as a finite disjoint union of sets in S.

We now wish to assign a 'correct' setting to talk about infinite coin tosses (or simple random walks on $\mathbb{Z}$). Let $\Omega = \{0,1\}^{\mathbb{N}}$, and we define

$$S = \{\{\mathbf{x} \mid x_1 = a_1, x_2 = a_2, \ldots, x_n = a_n\} \mid n \in \mathbb{N}, a_i \in \{0,1\}\} \cup \{\emptyset\} \tag{1.22}$$

and let $\mathcal{F} = \sigma(S)$. Verify that S is a π-system; we can set

$$\mu(\{\mathbf{x} \mid x_1 = a_1,\, x_2 = a_2, \ldots, x_n = a_n\}) = \prod_{i=1}^{n} p^{a_i}(1-p)^{1-a_i}. \tag{1.23}$$

For now, assume that μ is a pre-measure on S. Then there'll exist a unique probability measure on $(\Omega, \sigma(S))$ that extends μ.

1.3 Measurable Maps and Random Variables

Note that a space $(\Omega, \mathcal{F})$ is called a *measurable space*.

Definition 1.16. A map $T : (\Omega, \mathcal{F}) \to (\tilde{\Omega}, \tilde{\mathcal{F}})$ between two measurable spaces is termed a *measurable function* if $T^{-1}(A) \in \mathcal{F}$ for all $A \in \tilde{\mathcal{F}}$. When $(\tilde{\Omega}, \tilde{\mathcal{F}}) = (\mathbb{R}, \mathcal{B})$, we call T a *random variable*.

In most situations, X is usually a topological space and $\mathcal{B}(X)$ is the Borel σ-algebra on X. For the case of $X = \mathbb{R}$, a random variable can be interpreted as a meaningful quantity which can be studied.

Proposition 1.17. *$T : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B})$ is measurable if and only if $T^{-1}(U) \in \mathcal{F}$ whenever U is open in $\mathbb{R}$.*

Proof. One direction is clear; for the converse direction, let $\tilde{\mathcal{F}} = \{W \subseteq \mathbb{R} \mid T^{-1}(W) \in \mathcal{F}\}$. It is relatively easy to show that $\tilde{\mathcal{F}}$ is a σ-algebra on $\mathbb{R}$. Thus, since $\mathcal{B} \subseteq \tilde{\mathcal{F}}$, we have $T^{-1}(A) \in \mathcal{F}$ for all $A \in \mathcal{B}$. ■

Note that the image space being in $\mathbb{R}$ did not play a role in the above proof; for any image space $\tilde{\Omega}$, we can define $\tilde{\mathcal{F}} = \{W \subseteq \tilde{\Omega} \mid T^{-1}(W) \in \mathcal{F}\}$, and show that $\tilde{\mathcal{F}}$ is a σ-algebra on $\tilde{\Omega}$. Such a $\tilde{\mathcal{F}}$ is called the *pushforward σ-algebra* of $\mathcal{F}$ under T, and is the largest σ-algebra on $\tilde{\Omega}$ such that T is measurable. One can think in the converse way, where we are given a map $T : \Omega \to (\tilde{\Omega}, \tilde{F})$; to make such a T measurable, one can simply equip Ω with the powerset σ-algebra 2^{Ω}, and then T is automatically measurable. Alternatively, we can define $\mathcal{F} = \{T^{-1}(A) \mid A \in \tilde{\mathcal{F}}\}$, and then T is measurable. This is called the *pullback σ-algebra* of $\tilde{\mathcal{F}}$ under T, and is the smallest σ-algebra on Ω such that T is measurable.

The most non-trivial yet basic example of a random variable is the *indicator function* of a set $A \in \mathcal{F}$, which is defined as $\mathbf{1}_A(\omega) = 1$ if $\omega \in A$, and 0 otherwise. It is easy to check that $\mathbf{1}_A$ is a random variable.

Lemma 1.18. *If $X, Y : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B})$ are random variables, then $X + Y$ is also a random variable. More generally, if $f : \mathbb{R}^2 \to \mathbb{R}$ is continuous, then $f(X, Y)$ is a random variable.*

August 7th.

Example 1.19. Consider the infinite simple random walk from before; for $a \in \mathbb{Z}$, define $\tau_a = \inf\{n > 0 \mid S_n = a\}$, the first hitting time of a. Then we claim that τ_a is measurable and, hence, a random variable. To see this, note that $\tau_a^{-1}((n, \infty)) = \{\omega \in \Omega \mid \tau_a(\omega) > n\} = \{\omega \in \Omega \mid S_1(\omega) \neq a, S_2(\omega) \neq a, \ldots, S_n(\omega) \neq a\} = \bigcap_{k=1}^{n} \{\omega \in \Omega \mid S_k(\omega) \neq a\}$, which is in $\mathcal{F}$ since S_k is measurable for all k. Thus, τ_a is measurable.

Definition 1.20. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $T : (\Omega, \mathcal{F}) \to (\tilde{\Omega}, \tilde{\mathcal{F}})$ be a measurable map. Then, the *pushforward measure* of $\mathbb{P}$ under T is the measure $\tilde{\mathbb{P}}$ on $(\tilde{\Omega}, \tilde{\mathcal{F}})$ defined by $\tilde{\mathbb{P}}(A) = \mathbb{P}(T^{-1}(A))$ for all $A \in \tilde{\mathcal{F}}$.

Indeed, one can show that the pushforward measure $\tilde{\mathbb{P}}$ is a probability measure on $(\tilde{\Omega}, \tilde{\mathcal{F}})$.

Definition 1.21. Given a Borel probability measure μ on $(\mathbb{R}, \mathcal{B})$, we define the *cumulative distribution function* of μ to be the function $F_\mu : \mathbb{R} \to [0, 1]$ defined by $F_\mu(x) = \mu((-\infty, x])$ for all $x \in \mathbb{R}$.

Proposition 1.22. *With μ as above, the following hold true.*

1. *F_μ is non-decreasing.*
2. *$\lim_{x\to\infty} F_\mu(x) = 1$ and $\lim_{x\to-\infty} F_\mu(x) = 0$.*
3. *F_μ is right-continuous.*

Proof. 1. If $x_1 \leq x_2$, then $(-\infty, x_1] \subseteq (-\infty, x_2]$, and hence $F_\mu(x_1) = \mu((-\infty, x_1]) \leq \mu((-\infty, x_2]) = F_\mu(x_2)$.

2. Note that $\mathbb{R} = \bigcup_{n=1}^{\infty}(-\infty, n]$; defining $A_k = (-\infty, k]$, we have $A_1 \subseteq A_2 \subseteq \cdots$ and $A_k \uparrow \mathbb{R}$. Since μ is a measure, $\mu(A_k) \uparrow \mu(\mathbb{R}) = 1$, and hence $\lim_{x\to\infty} F_\mu(x) = 1$. Similarly, $\lim_{x\to-\infty} F_\mu(x) = 0$.

3. Let x_n be a decreasing sequence of numbers such that $x_n \downarrow x$. Then, $(-\infty, x_1] \supseteq (-\infty, x_2] \supseteq \cdots$ and $(-\infty, x_n] \downarrow (-\infty, x]$. Since μ is a measure, $\mu((-\infty, x_n]) \downarrow \mu((-\infty, x])$, and hence $F_\mu(x_n) \downarrow F_\mu(x)$. Thus, F_μ is right-continuous.

■

Theorem 1.23. *Suppose $F : \mathbb{R} \to [0, 1]$ is a non-decreasing, right-continuous function such that $\lim_{x\to-\infty} F(x) = 0$ and $\lim_{x\to\infty} F(x) = 1$. Then, there exists a unique Borel probability measure μ on $(\mathbb{R}, \mathcal{B})$ such that $F = F_\mu$.*

Thus, there is a one-to-one correspondence between Borel probability measures on $(\mathbb{R}, \mathcal{B})$ and cumulative distribution functions. In general, it easier to study the cumulative distribution function of a probability measure than the measure itself.

1.3.1 The Pushfoward Measure

August 11th.

In many situations, we can model a random experiment in several ways. Let $\Omega = [0, 1]$, $\mathcal{F} = \mathcal{B}$, and λ the Lebesgue measure. Define the random variables $X_1 = \mathbf{1}_{[0,1/2]}$ and $X_2 = \mathbf{1}_{[0,1/4]\cup[1/2,3/4]}$. Consider $\Gamma = (X_1, X_2) : \Omega \to \mathbb{R}^2$, and let $\mu = \lambda \circ \Gamma^{-1}$ be the pushforward measure of λ under Γ. The support of μ is the set $\Gamma(\Omega) = \{(1, 0), (0, 1), (1, 1), (0, 0)\}$. The measure at each of these points is given by $\mu(\{(1, 0)\}) = \lambda(\Gamma^{-1}(\{(1, 0)\})) = \lambda([0, 1/4]) = 1/4$, $\mu(\{(0, 1)\}) = \lambda(\Gamma^{-1}(\{(0, 1)\})) = \lambda([1/2, 3/4]) = 1/4$, $\mu(\{(1, 1)\}) = \lambda(\Gamma^{-1}(\{(1, 1)\})) = \lambda([1/4, 1/2]) = 1/4$, and $\mu(\{(0, 0)\}) = \lambda(\Gamma^{-1}(\{(0, 0)\})) = \lambda([3/4, 1]) = 1/4$. Thus, we have a probability measure on the support of μ. That is, we have modeled tossing a fair coin twice, without explicitly defining the sample space as $\{0, 1\}^2$. The probability space does not matter in most contexts; the random variables and their distributions are what matter.

Naturally, we can ask further questions. Whether the following models can exist: tossing a coin n times, and tossing a coin countably many times. The first question is easy to answer; we can define the k-th random variable X_k exactly as

$$X_k = \mathbf{1}_{[0,\frac{1}{2^k}]\cup[\frac{2}{2^k},\frac{3}{2^k}]\cup\cdots\cup[\frac{2^k-2}{2^k},\frac{2^k-1}{2^k}]}. \tag{1.24}$$

We leave it as an exercise to verify that $(X_1, X_2, \ldots, X_n)$ has the same distribution as tossing a fair coin n times.

Theorem 1.24. *Given any probability measure μ on $(\mathbb{R}, \mathcal{B})$, there exists a measurable map $T : ([0, 1], \mathcal{B}([0, 1]), \lambda) \to (\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ such that $\mu = \lambda \circ T^{-1}$. That is, the pushforward measure of λ under T is μ.*

Before we proceed with this theorem, we show the following proposition.

Proposition 1.25. *For any Borel probability measures μ and ν on $(\mathbb{R}, \mathcal{B})$, if $F_\mu = F_\nu$, then $\mu = \nu$. That is, the cumulative distribution function uniquely determines the probability measure.*

Proof. Let μ and ν be the required measures such that $F_\mu = F_\nu$, or $\mu((-\infty, x]) = \nu((-\infty, x])$ for all $x \in \mathbb{R}$. Let $S = \{(-\infty, x] \mid x \in \mathbb{R}\}$, which is a π-system. Note that $\sigma(S) = \mathcal{B}$; thus, by the Sierpiński-Dynkin π-λ theorem, we have $\mu(A) = \nu(A)$ for all $A \in \mathcal{B}$. Hence, $\mu = \nu$. ■

We now prove **Theorem 1.23** and **Theorem 1.24** together.

Proof of theorems. Suppose $F : \mathbb{R} \to [0,1]$ be a non-decreasing, right-continuous function such that $\lim_{x\to-\infty} F(x) = 0$ and $\lim_{x\to\infty} F(x) = 1$. We will show that there exists a measurable map $T : ([0,1], \mathcal{B}([0,1]), \lambda) \to (\mathbb{R}, \mathcal{B})$ such that $F_{\lambda\circ T^{-1}} = F$; note that this implies the second theorem by the previous proposition.

For such an F, we can define a generalized inverse T defined as $T(x) = \inf\{p \in \mathbb{R} \mid F(p) \geq x\}$. For any $x \in [0,1]$, the set $\{p \in \mathbb{R} \mid F(p) \geq x\}$ is non-empty (since $\lim_{p\to\infty} F(p) = 1$) and has infimum, so T is well-defined. Moreover, T is non-decreasing and takes values in $\mathbb{R}$. To show measurability of T, note that for any $a \in \mathbb{R}$, we have

$$\{x \in [0,1] \mid T(x) > a\} = \{x \in [0,1] \mid F(a) < x\} = (F(a), 1] \cap [0,1], \tag{1.25}$$

which is a Borel set in $[0,1]$. Thus, T is measurable. We now verify that $F_{\lambda\circ T^{-1}} = F$; For any $x \in \mathbb{R}$, we have

$$\begin{aligned}(\lambda \circ T^{-1})((-\infty, x]) &= \lambda(T^{-1}((-\infty, x])) = \lambda(\{u \in [0,1] \mid T(u) \leq x\}) \\ &= \lambda(\{u \in [0,1] \mid \inf\{p \in \mathbb{R} \mid F(p) \geq u\} \leq x\}) = \lambda(\{u \in [0,1] \mid F(x) \geq u\}) \\ &= \lambda([0, F(x)]) = F(x). \end{aligned} \tag{1.26}$$

Thus, $F_{\lambda\circ T^{-1}} = F$, establishing the existence of the desired measure. ■

1.4 Probability Measures on $\mathbb{R}$

August 14th.

We first wish to generalize the notion of the integral of a function with respect to the Lebesgue measure. A function $\varphi : [a,b] \to \mathbb{R}$ is said to be a *simple function* if it is of the form $\varphi = \sum_{i=1}^n a_i \mathbf{1}_{A_i}$ for some $n \in \mathbb{N}$, $a_i \in \mathbb{R}$, and $A_i \in \mathcal{B}$. That is, φ takes finitely many values, and the pre-image of each value is a Borel set. For a simple function φ, we define the integral of φ to be

$$\int_{[a,b]} \varphi(x)\, dx = \sum_{i=1}^n a_i \lambda(A_i). \tag{1.27}$$

In general, a function $g : [a,b] \to \mathbb{R}$ is said to be *Borel measurable* if $g^{-1}(A) \in \mathcal{B}$ for all Borel sets $A \subseteq \mathbb{R}$. Note that any simple function is Borel measurable. The integral of a non-negative Borel measurable function $g : [a,b] \to \mathbb{R}$ is then defined as

$$\int_{[a,b]} g(x)\, dx = \sup\left\{ \int_{[a,b]} \varphi(x)\, dx \;\middle|\; 0 \leq \varphi \leq g,\ \varphi \text{ is simple} \right\}. \tag{1.28}$$

One can infer that $\int(f + \alpha g) = \int f + \alpha \int g$, $g \leq f \implies \int g \leq \int f$, and $\int_{I_1 \sqcup I_2} g = \int_{I_1} g + \int_{I_2} g$ for disjoint intervals I_1 and I_2, and non-negative Borel measurable functions f and g. We can extend the definition of the integral to all Borel measurable functions g by defining $\int g = \int g^+ - \int g^-$, where $g^+ = \max\{g, 0\}$ and $g^- = \max\{-g, 0\}$. Thus, we are fit to define the following.

Definition 1.26. A probability measure $\mu \in \mathcal{P}(\mathbb{R})$ is said to have a *density* f if $f : \mathbb{R} \to \mathbb{R}$ is a non-negative Borel measurable function such that $\int_{\mathbb{R}} f(x)\, dx = 1$ and $F_\mu(x) = \int_{-\infty}^x f(t)\, dt$ for all $x \in \mathbb{R}$.

Suppose $\mu \in \mathcal{P}(\mathbb{R})$ has a density f. Then, for any Borel set $A \subseteq \mathbb{R}$, we have

$$\mu(A) = \int_A f(x)\, dx. \tag{1.29}$$

Note that not every probability measure has a density; for example, the measure $\mu = \delta_0$, indicator of the set $\{0\}$, does not have a density. In fact, if there exists $p \in \mathbb{R}$ with $\mu(\{p\}) > 0$, then μ does not have a

density. This is because if μ has a density f, then $\mu(\{p\}) = \int_{\{p\}} f(x)\,dx = 0$, which is a contradiction. Thus, any probability measure with a density must be non-atomic; that is, it cannot assign positive measure to any singleton set. However, the converse is not true; there exist non-atomic probability measures that do not have a density. For example, consider the Cantor measure μ on $[0,1]$, which is non-atomic, but does not have a density.

We now wish to define a metric on the space of probability measures on $\mathbb{R}$.

1. For any $\mu, \nu \in \mathcal{P}(\mathbb{R})$, we define the *total variation distance* between μ and ν to be
$$D_1(\mu,\nu) = d_{TV}(\mu,\nu) := \sup_{A\in\mathcal{B}} |\mu(A) - \nu(A)|\,. \tag{1.30}$$

2. The CDFs may be utilized to define
$$D_2(\mu,\nu) := \sup_{x\in\mathbb{R}} |F_\mu(x) - F_\nu(x)|\,. \tag{1.31}$$

Both of these metrics are fine as they are, but fail in being 'nice' with respect to convergence of measures; the sequence of measures $\{\mu_n = \delta_{1/n}\}_{n=1}^\infty$ fails to converge to δ_0 in either of these metrics, even though we expect it to converge intuitively.

August 18th.

Hereforth, we define the *Levy distance* on $\mathcal{P}(\mathbb{R})$ as
$$d(\mu,\nu) := \inf\{\epsilon > 0 \mid F_\mu(t+\epsilon) + \epsilon \geq F_\nu(t),\ F_\nu(t+\epsilon) + \epsilon \geq F_\mu(t) \text{ for all } t \in \mathbb{R}\}. \tag{1.32}$$

In fact, this can be generalized to a metric on Borel probability measures on any metric space $(X, \mathcal{D})$ as follows (this is termed the *Levy-Prokhorov metric*).
$$d_{LP}(\mu,\nu) := \inf\{\epsilon > 0 \mid \mu(A+\epsilon) + \epsilon \geq \nu(A),\ \nu(A+\epsilon) + \epsilon \geq \mu(A) \text{ for all closed } A \subseteq X\} \tag{1.33}$$

where the ϵ-*fattening* of A is defined as
$$A + \epsilon := \{x \in X \mid \text{ there exists } y \in A \text{ such that } \mathcal{D}(x,y) < \epsilon\}. \tag{1.34}$$

Note that these two metrics are not the same on $\mathcal{P}(\mathbb{R})$, but are equivalent. Also, via the Levy distance, we can show $d(\delta_{1/n}, \delta_0) = 1/n$, showing that $\delta_{1/n} \to \delta_0$ in this metric.

August 25th.

Theorem 1.27. *Suppose $\mu, \mu_1, \mu_2, \ldots \in \mathcal{P}(\mathbb{R})$ are Borel probability measures. Then $d(\mu_n, \mu) \xrightarrow{n\to\infty} 0$ if and only if $F_{\mu_n}(x) \xrightarrow{n\to\infty} F_\mu(x)$ for all continuity points $x \in \mathbb{R}$ of F_μ.*

Proof. We first show the forward implication. Suppose $d(\mu_n, \mu) \to 0$. Then, for any $\epsilon > 0$, there exists $N_\epsilon \in \mathbb{N}$ such that $d(\mu_n, \mu) < \epsilon$ for all $n \geq N_\epsilon$. Now fix $\epsilon > 0$ and $x \in \mathbb{R}$. Then, for all $n \geq N_\epsilon$, we have $F_\mu(x+\epsilon) + \epsilon \geq F_{\mu_n}(x)$. We can replace the right hand side with the limit superior to get
$$F_\mu(x+\epsilon) + \epsilon \geq \limsup_{n\to\infty} F_{\mu_n}(x) \quad \text{for all } \epsilon > 0. \tag{1.35}$$

This implies that $F_\mu(x) \geq \limsup_{n\to\infty} F_{\mu_n}(x)$. Similarly, for all $n \geq N_\epsilon$, we have $F_{\mu_n}(x) + \epsilon \geq F_\mu(x - \epsilon)$. By a similar argument, we have $\liminf_{n\to\infty} F_{\mu_n}(x) \geq F_\mu(x)$ if x is a continuity point of F_μ. Thus, we have shown that $\limsup_{n\to\infty} F_{\mu_n}(x) \leq F_\mu(x) \leq \liminf_{n\to\infty} F_{\mu_n}(x)$ or $F_{\mu_n}(x) \to F_\mu(x)$ for all continuity points.

For the converse, if $\mathcal{C}$ denotes the set of continuity points of F_μ, then $\mathcal{C}$ is cocountable and hence dense in $\mathbb{R}$. Now fix $\varepsilon \in (0,1)$. Let $A, B \in \mathbb{R}$ be numbers such that $F_\mu(A) < \varepsilon$ and $F_\mu(B) \geq 1 - \varepsilon$. Pick points $\{x_i\}_{i=0}^m$ in $[A,B] \cap \mathcal{C}$ such that $A = x_0 < x_1 < \cdots < x_m = B$ and $x_{i+1} - x_i < \varepsilon$ for all i. By the choice of the x_i's, we have $F_{\mu_n}(x_i) \to F_\mu(x_i)$ for all i, and so there exists $N_\varepsilon \in \mathbb{N}$ such that $|F_{\mu_n}(x_i) - F_\mu(x_i)| < \varepsilon$ for all $n \geq N_\varepsilon$ and all i. Now fix $t \in [A,B]$; so $t \in [x_i, x_{i+1}]$ for some i. Then, for all $n \geq N_\varepsilon$, we have
$$F_{\mu_n}(t+\varepsilon) \geq F_{\mu_n}(x_{i+1}) \geq F_\mu(x_{i+1}) - \varepsilon \geq F_\mu(t) - \varepsilon \implies F_{\mu_n}(t+\varepsilon) + \varepsilon \geq F_\mu(t) \tag{1.36}$$

for all $n \geq N_\varepsilon$ and $t \in [A, B]$. If $t > B$, then both $F_{\mu_n}(t)$ and $F_\mu(t)$ are at least $1 - 2\varepsilon$, and hence $F_{\mu_n}(t+\varepsilon)+\varepsilon \geq F_\mu(t)$. If $t < A$, then both $F_{\mu_n}(t)$ and $F_\mu(t)$ are at most 2ε, and hence $F_{\mu_n}(t+\varepsilon)+\varepsilon \geq F_\mu(t)$. Thus, we have shown that for all $\varepsilon > 0$, there exists $N_\varepsilon \in \mathbb{N}$ such that for all $n \geq N_\varepsilon$, we have

$$F_{\mu_n}(t+\varepsilon) + \varepsilon \geq F_\mu(t) \quad \text{for all } t \in \mathbb{R} \implies d(\mu_n, \mu) \leq 2\varepsilon. \tag{1.37}$$

■

1.5 Expectation

Expectation, in our new setting, corresponds to integration in a probability space. To this end, we start with simple functions.

Definition 1.28. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A measurable function $S : \Omega \to \mathbb{R}$ is called a *simple random variable* if it takes finitely many values; that is, $S(\Omega) = \{s_1, \ldots, s_n\}$ for some $n \in \mathbb{N}$.

In this case, we can write $S = \sum_{i=1}^n c_i \mathbf{1}_{A_i}$ for some $c_i \in \mathbb{R}$ and $A_i \in \mathcal{F}$. This representation need not be unique; however, note that any such simple function can be written in a unique way as

$$S = \sum_{i=1}^n \beta_i \mathbf{1}_{B_i} \tag{1.38}$$

where $\beta_i \in S(\Omega)$ are unequal and B_i's are measurable sets that are mutually disjoint.

Definition 1.29. The *expectation of a simple random variable* $S = \sum_{i=1}^n c_i \mathbf{1}_{A_i}$ is defined as

$$\mathbb{E}[S] := \sum_{i=1}^n c_i \mathbb{P}(A_i). \tag{1.39}$$

Since this representation is not unique, one can verify that the expectation is well-defined by using the unique representation of S as $\sum_{i=1}^n \beta_i \mathbf{1}_{B_i}$. Also, if S and T are simple random variables, then $S+T$ is also a simple random variable, and $\mathbb{E}[S+T] = \mathbb{E}[S] + \mathbb{E}[T]$. Similarly, for any $\alpha \in \mathbb{R}$, we have $\mathbb{E}[\alpha S] = \alpha \mathbb{E}[S]$. If $S \geq 0$, then $\mathbb{E}[S] \geq 0$. We can now extend the definition of expectation to all non-negative random variables, and then to all random variables.

Definition 1.30. The *expectation of a non-negative random variable* $X : \Omega \to [0, \infty)$ is defined as

$$\mathbb{E}[X] := \sup\{\mathbb{E}[S] \mid 0 \leq S \leq X, S \text{ is a simple random variable}\}. \tag{1.40}$$

If X is a general random variable, we can write $X = X^+ - X^-$, where $X^+ = \max\{X, 0\}$ and $X^- = \max\{-X, 0\}$. If $\mathbb{E}[X^+] < \infty$ or $\mathbb{E}[X^-] < \infty$, then we define the *expectation of (the) random variable* X as

$$\mathbb{E}[X] := \mathbb{E}[X^+] - \mathbb{E}[X^-]. \tag{1.41}$$

Proposition 1.31. *Let X and Y be non-negative random variables. Then, the following hold true.*

1. $\mathbb{E}[X] \geq 0$.
2. $\mathbb{E}[X+Y] = \mathbb{E}[X] + \mathbb{E}[Y]$, *if at least one of* $\mathbb{E}[X]$ *or* $\mathbb{E}[Y]$ *is finite.*

Proof. The first part is clear from the definition since $S \equiv 0$ is a simple random variable such that $0 \leq S \leq X$, so $\mathbb{E}[X] \geq \mathbb{E}[S] = 0$. For the second part, let S and T be simple random variables such that $0 \leq S \leq X$ and $0 \leq T \leq Y$. Then, $S+T$ is a simple random variable such that $0 \leq S+T \leq X+Y$, and hence $\mathbb{E}[X+Y] \geq \mathbb{E}[S+T] = \mathbb{E}[S] + \mathbb{E}[T]$. Taking supremum over all such S and T, we get $\mathbb{E}[X+Y] \geq \mathbb{E}[X] + \mathbb{E}[Y]$. For the reverse inequality, let U be a simple random variable such that $0 \leq U \leq X+Y$. Then, we can write $U = U_1 + U_2$ where $U_1 = U\mathbf{1}_{\{X>0\}}$ and $U_2 = U\mathbf{1}_{\{Y>0\}}$. Then,

$0 \leq U_1 \leq X$ and $0 \leq U_2 \leq Y$, and hence $\mathbb{E}[U] = \mathbb{E}[U_1] + \mathbb{E}[U_2] \leq \mathbb{E}[X] + \mathbb{E}[Y]$. Taking supremum over all such U, we get $\mathbb{E}[X+Y] \leq \mathbb{E}[X] + \mathbb{E}[Y]$. Thus, we have equality. ■

1.5.1 Results of Convergence

August 28th.

We also have the monotone converge theorem, of which we state a basic version first.

Theorem 1.32. *Suppose $X \geq 0$ is a random variable, and $\{\varphi_n\}_{n=1}^{\infty}$ is a sequence of simple random variables such that $0 \leq \varphi_n \uparrow X$. Then, $\mathbb{E}[\varphi_n] \uparrow \mathbb{E}[X]$.*

Proof. Fix $\varepsilon > 0$. By definition, there exists a simple function S such that $0 \leq S \leq X$ and $\mathbb{E}[S] \geq \mathbb{E}[X] - \varepsilon$. For a fixed $\omega \in \Omega$, there exists a natural $N_\omega \in \mathbb{N}$ such that $\varphi_n(\omega) \geq S(\omega) - \varepsilon$ for all $n \geq N_\omega$. Now define $A_n = \{\omega \in \Omega \mid \varphi_n(\omega) \geq S(\omega) - \varepsilon\}$; since φ_n's are increasing functions, we have $A_1 \subseteq A_2 \subseteq \cdots \subseteq A_n \subseteq A_{n+1} \subseteq \cdots$, and $\bigcup_{n=1}^{\infty} A_n = \Omega$. Thus, given $\delta > 0$, we can find a natural $N_\delta \equiv N$ such that $\mathbb{P}(A_{N_\delta}) \geq 1 - \delta$. Now, for all $n \geq N$, we have

$$\mathbb{E}[\varphi_n] = \mathbb{E}[\varphi_n \mathbf{1}_{A_N}] + \mathbb{E}[\varphi_n \mathbf{1}_{A_N^c}]. \tag{1.42}$$

But note that on such an A_N, we have $\varphi_n \geq S - \varepsilon$, and hence $\mathbb{E}[\varphi_n \mathbf{1}_{A_N}] \geq \mathbb{E}[(S-\varepsilon)\mathbf{1}_{A_N}] = \mathbb{E}[S\mathbf{1}_{A_N}] - \varepsilon\mathbb{P}(A_N)$. If $M = \sup_{\omega\in\Omega} S(\omega)$, then $\mathbb{E}[\varphi_n \mathbf{1}_{A_N^c}] \geq 0$ and $\mathbb{E}[S\mathbf{1}_{A_N}] = \mathbb{E}[S] - \mathbb{E}[S\mathbf{1}_{A_N^c}] \geq \mathbb{E}[S] - M\mathbb{P}(A_N^c)$. Thus, we have

$$\mathbb{E}[\varphi_n] \geq \mathbb{E}[S] - M\mathbb{P}(A_N^c) - \varepsilon\mathbb{P}(A_N) \geq \mathbb{E}[S] - M\delta - \varepsilon. \tag{1.43}$$

Choosing $\delta = \varepsilon/M$, we have $\mathbb{E}[\varphi_n] \geq \mathbb{E}[S] - 2\varepsilon \geq \mathbb{E}[X] - 3\varepsilon$. Since $\varepsilon > 0$ was arbitrary, we have $\lim_{n\to\infty} \mathbb{E}[\varphi_n] = \mathbb{E}[X]$. ■

Theorem 1.33 (The *monotone convergence theorem*)**.** *Suppose $X \geq 0$ is a random variable, and $\{X_n\}_{n=1}^{\infty}$ is a sequence of non-negative random variables such that $0 \leq X_n \uparrow X$. Then, $\mathbb{E}[X_n] \uparrow \mathbb{E}[X]$.*

Lemma 1.34 (*Fatou's lemma*)**.** *Suppose $\{X_n\}_{n=1}^{\infty}$ is a sequence of non-negative random variables. Then,*

$$\mathbb{E}[\liminf_{n\to\infty} X_n] \leq \liminf_{n\to\infty} \mathbb{E}[X_n]. \tag{1.44}$$

Proof. Define $Y_n = \inf_{k\geq n} X_k$ for all $n \in \mathbb{N}$. Then, $\{Y_n\}_{n=1}^{\infty}$ is a sequence of non-negative random variables such that $Y_1 \leq Y_2 \leq \cdots$ and $Y_n \uparrow \liminf_{n\to\infty} X_n$. By the monotone convergence theorem, we have

$$\mathbb{E}[\liminf_{n\to\infty} X_n] = \lim_{n\to\infty} \mathbb{E}[Y_n] \leq \liminf_{n\to\infty} \mathbb{E}[X_n], \tag{1.45}$$

where the last inequality follows from the fact that $Y_n \leq X_n$ for all n. ■

September 1st.

Theorem 1.35 (The *dominated convergence theorem*)**.** *Suppose $\{X_n\}_{n=1}^{\infty}$ is a sequence of random variables such that $X_n \to X$ almost surely, and there exists a non-negative random variable Y such that $|X_n| \leq Y$ for all n, and $\mathbb{E}[Y] < \infty$. Then, $\mathbb{E}[X_n] \to \mathbb{E}[X]$.*

Proof. Define $Z_n = 2Y - |X_n - X|$. Since $\mathbb{Z}_n \geq 0$, we can apply Fatou's lemma to get

$$\mathbb{E}[\liminf_{n\to\infty} Z_n] \leq \liminf_{n\to\infty} \mathbb{E}[Z_n] \implies 2\mathbb{E}[Y] - \mathbb{E}[\limsup_{n\to\infty} |X_n - X|] \leq 2\mathbb{E}[Y] - \limsup_{n\to\infty} \mathbb{E}[|X_n - X|], \tag{1.46}$$

which implies that $\mathbb{E}[\limsup_{n\to\infty} |X_n - X|] \geq \limsup_{n\to\infty} \mathbb{E}[|X_n - X|]$. Now, since $X_n \to X$ almost surely, we have $\limsup_{n\to\infty} |X_n - X| = 0$ almost surely, and hence $\mathbb{E}[\limsup_{n\to\infty} |X_n - X|] = 0$. Thus, we have $\limsup_{n\to\infty} \mathbb{E}[|X_n - X|] = 0$, which implies that $\mathbb{E}[|X_n - X|] \to 0$, and hence $\mathbb{E}[X_n] \to \mathbb{E}[X]$. ■

Chapter 2

MORE ON RANDOM VARIABLES

2.1 Independence

September 15th.

Definition 2.1. Let $(\Omega, \mathcal{F}, P)$ be a probability space.

1. We say sub-σ-algebras $\mathcal{G}_1, \mathcal{G}_2, \cdots, \mathcal{G}_n$ of $\mathcal{F}$ are independent if $\mathbb{P}(A_1 \cap A_2 \cap \cdots \cap A_n) = \mathbb{P}(A_1)\mathbb{P}(A_2)\ldots\mathbb{P}(A_n)$ for all $A_i \in \mathcal{G}_i$, $i = 1, 2, \ldots, n$.
2. We say that the random variables $X_1, X_2, \ldots, X_n$ are independent if the sub-σ-algebras $\sigma(X_1), \sigma(X_2), \ldots, \sigma(X_n)$ are independent.
3. We say that the events $A_1, A_2, \ldots, A_n$ are independent if the sub-σ-algebras $\sigma(A_1), \sigma(A_2), \ldots, \sigma(A_n)$ are independent.

Equivalent condition for independence of events is that

$$\mathbb{P}(A_1^{\pm} \cap A_2^{\pm} \cap \cdots \cap A_n^{\pm}) = \mathbb{P}(A_1^{\pm})\mathbb{P}(A_2^{\pm})\cdots\mathbb{P}(A_n^{\pm}), \tag{2.1}$$

where $A_i^+ = A_i$ and $A_i^- = A_i^c$ for $i = 1, 2, \ldots, n$.

Definition 2.2. We say that a collection of countably many random variables $X_1, X_2, \ldots$ on $(\Omega, \mathcal{F}, \mathbb{P})$ are independent if every finite subcollection of them are independent.

Definition 2.3. Suppose $X_1, X_2, \ldots$ are random variables on $(\Omega, \mathcal{F}, \mathbb{P})$, and define $\mathcal{F}_n = \sigma(X_n, X_{n+1}, \ldots)$ for $n = 1, 2, \ldots$. The *tail σ-algebra* of the sequence $X_1, X_2, \ldots$ is defined as

$$\mathcal{T} = \bigcap_{n=1}^{\infty} \mathcal{F}_n \subseteq \sigma(X_1, X_2, \ldots) = \mathcal{F}. \tag{2.2}$$

An event $A \in \mathcal{T}$ is an event whose occurrence is determined by the variables $X_n, X_{n+1}, \ldots$ for every n. Thus, A does not depend on any fixed finite number of the variables at the beginning of the sequence. For example, the event

$$A = \{X_n \text{ occurs infinitely often}\} \tag{2.3}$$

belongs to $\mathcal{T}$, since for every m it can be described using only $X_m, X_{m+1}, \ldots$.

If $X_1, X_2, \ldots$ are independent and $A \in \mathcal{T}$, then A is independent of $\sigma(X_1, \ldots, X_n)$ for every n, since A is measurable with respect to $\sigma(X_{n+1}, X_{n+2}, \ldots)$. Therefore A is independent of

$$\sigma(X_1, X_2, \ldots) = \sigma\left(\bigcup_{n=1}^{\infty} \sigma(X_1, \ldots, X_n)\right). \tag{2.4}$$

In particular, A is independent of itself, and hence

$$\mathbb{P}(A) = \mathbb{P}(A \cap A) = \mathbb{P}(A)^2. \tag{2.5}$$

Thus $\mathbb{P}(A) \in \{0, 1\}$. This is known as *Kolmogorov's zero–one law*, that for independent random variables, any tail event has probability 0 or 1. Without independence, a tail event may retain information about the variables in every finite initial segment, and this argument does not apply.

2.1.1 Percolation

September 18th.

We first discuss (Bernoulli) percolation on $\mathbb{Z}^2 \subseteq \mathbb{R}^2$. To each edge e, we associate $X_e \sim \text{Ber}(p)$ for some $p \in [0, 1]$, independently for each edge. If $X_e = 1$, we retain the edge e, and if $X_e = 0$, we delete the edge e. We then have a random subgraph of $\mathbb{Z}^2$, which is a random subset of edges of $\mathbb{Z}^2$. The present or retained edges are termed open, and the absent or deleted edges are termed closed. We ask if there exists an unbounded connected component of open edges; let E_p be the event that there exists an unbounded connected component of open edges. Since there are only countably many X_e's, we can order them in a spiral like manner starting from the origin, and let $X_1, X_2, \ldots$ be the corresponding sequence of random variables. Then E_p is a tail event here, and hence by Kolmogorov's zero–one law, $\mathbb{P}(E_p) \in \{0, 1\}$. Moreover, $\mathbb{P}(E_0) = 0$ and $\mathbb{P}(E_1) = 1$, and for $0 < p < q < 1$, we have $\mathbb{P}(E_p) \leq \mathbb{P}(E_q)$ by a coupling argument. Therefore, there exists a critical probability $p_c \in [0, 1]$ such that $\mathbb{P}(E_p) = 0$ for $p < p_c$ and $\mathbb{P}(E_p) = 1$ for $p > p_c$.

Let $\{0 \leftrightarrow \infty\}$ denote the event that there exists an unbounded connected component of open edges containing the origin. Define $\theta_p := \mathbb{P}(0 \leftrightarrow \infty)$. Note that this is not a tail event, but $\theta_p > 0$ if and only if $\mathbb{P}(E_p) = 1$. Also, θ_p is non-decreasing in p.

Lemma 2.4. *For small p, $\mathbb{P}(E_p) = 0$. For $p \approx 1$, $\mathbb{P}(E_p) = 1$.*

Proof. Firstly, note that the event $\{0 \leftrightarrow \infty\}$ is contained in the event that there exists a self avoiding path of open edges starting from the origin and ending at manhattan distance N from the origin, for some $N \in \mathbb{N}$. This event is further contained in the event that there exists a self avoiding path of open edges starting from the origin with length at least N. Call the first event A_N and the second event B_N. Then $\{0 \leftrightarrow \infty\} \subseteq A_N \subseteq B_N$, and hence $\mathbb{P}(0 \leftrightarrow \infty) \leq \mathbb{P}(A_N) \leq \mathbb{P}(B_N)$. If C_n is the event that there exists a self avoiding path of open edges starting from the origin of length exactly N, then $\mathbb{P}(C_N) \leq 4 \cdot 3^{N-1} p^N \leq (4p)^N$, since there are at most $4 \cdot 3^{N-1}$ self avoiding paths of length N starting from the origin, and each such path is open with probability p^N. Moreover, note that $B_N = \bigcup_{n=N}^{\infty} C_n$, and hence $\mathbb{P}(B_N) \leq \sum_{n=N}^{\infty} (4p)^n$. Therefore, if $p < 1/4$, we have $\mathbb{P}(B_N) = \frac{(4p)^N}{1-4p} \to 0$ as $N \to \infty$, and hence $\mathbb{P}(0 \leftrightarrow \infty) = 0$ for $p < 1/4$. So, for small p, specifically for $p < 1/4$, we have $\theta_p = 0$ and hence $\mathbb{P}(E_p) = 0$.

For the second statement, note that the $\text{Ber}(p)$ percolation model on $\mathbb{Z}^2$ induces a $\text{Ber}(1-p)$ percolation model on the dual lattice $(\mathbb{Z}^2)^*$. Thus, the event $\{0 \not\leftrightarrow \infty\}$ is exactly the event that there exists a loop of open edges in the dual lattice surrounding the origin. If C_N^* is the event that there exists a loop of open edges in the dual lattice surrounding the origin with length exactly N, then this event is contained in the event that there exists a self avoiding path of open edges in the dual lattice starting at $\{(1, 0), (2, 0), \ldots, (n, 0)\}$ of length exactly N. Therefore, $\mathbb{P}(C_N^*) \leq n(4(1-p))^N$, and hence $\mathbb{P}(\{0 \not\leftrightarrow \infty\}) \leq \sum_{N=1}^{\infty} n(4(1-p))^N$. Therefore, if $p > 3/4$, we have $\mathbb{P}(\{0 \not\leftrightarrow \infty\}) = 0$, and hence $\theta_p = 1$ and $\mathbb{P}(E_p) = 1$ for $p > 3/4$. ∎

Lemma 2.5 (*Borel-Cantelli lemma*)**.** *Let $A_1, A_2, \ldots$ be a sequence of events in a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.*

1. *If $\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$, then $\mathbb{P}(A_n \text{ i.o.}) = 0$.*
2. *If $A_1, A_2, \ldots$ are independent and $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$, then $\mathbb{P}(A_n \text{ i.o.}) = 1$.*

We prove the second statement.

Proof. Let $B_n = \bigcup_{j=n}^{\infty} A_j$, so that $B_1 \supseteq B_2 \supseteq \cdots$ and $\mathbb{P}(A_n \text{ i.o.}) = \mathbb{P}(\bigcap_{n=1}^{\infty} B_n)$. If we prove that $\mathbb{P}(B_n) = 1$ for all n, then we are done. Now, $\mathbb{P}(B_n) = 1 - \mathbb{P}(B_n^c) = 1 - \mathbb{P}(\bigcap_{j=n}^{\infty} A_j^c)$. Since $A_1, A_2, \ldots$ are

independent, we have

$$\mathbb{P}(\bigcap_{j=n}^{m} A_j^c) = \prod_{j=n}^{m} \mathbb{P}(A_j^c) = \prod_{j=n}^{m} (1 - \mathbb{P}(A_j)) \leq \prod_{j=n}^{m} e^{-\mathbb{P}(A_j)} = e^{-\sum_{j=n}^{m} \mathbb{P}(A_j)}. \tag{2.6}$$

Taking the limit as $m \to \infty$, we get $\mathbb{P}(\bigcap_{j=n}^{\infty} A_j^c) \leq e^{-\sum_{j=n}^{\infty} \mathbb{P}(A_j)} = 0$, since $\sum_{j=n}^{\infty} \mathbb{P}(A_j) = \infty$. Therefore, $\mathbb{P}(B_n) = 1$ for all n, and hence $\mathbb{P}(A_n \text{ i.o.}) = 1$. ■

2.2 Moments

For a random variable $X : (\Omega, \mathcal{F}, \mathbb{P}) \to \mathbb{R}$, recall that the *mean* of X was simply its expectation $\mathbb{E}[X]$, if it exists. Else, the mean is undefined. For higher moments, the *p-th moment* of X is defined as $\mathbb{E}[|X|^p]$, if it exists. The *variance* of X is defined as $\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$.

In many cases, for a given X, we may not know its exact distribution, but we may know its moments. In such cases, we can use the moments to bound the probabilities of certain events. For example, if X is a non-negative random variable with finite mean $\mathbb{E}[X]$, then by Markov's inequality, for any $t > 0$, we have

$$\mathbb{P}(X > t) \leq \frac{\mathbb{E}[X]}{t}. \tag{2.7}$$

In particular, if $\mathbb{E}[X] = 0$, then $\mathbb{P}(X \geq t) = 0$ for all $t > 0$, and hence $X = 0$ almost surely. If X also happens to have a finite second moment $\mathbb{E}[X^2]$, then we can replace X with $|X - \mathbb{E}[X]|$ in the above inequality to get Chebyshev's inequality, which states that for any $t > 0$,

$$\mathbb{P}(|X - \mathbb{E}[X]| > t) \leq \frac{\mathrm{Var}(X)}{t^2}. \tag{2.8}$$

In fact, if $\mathbb{E}[|X - \mathbb{E}[X]|^n] < \infty$, then

$$\mathbb{P}(|X - \mu| > t) \leq \frac{\mathbb{E}[|X - \mu|^n]}{t^n}. \tag{2.9}$$

The *exponential moment* of X is defined as $\mathbb{E}[e^{\lambda x}]$ for $\lambda \neq 0$, if it exists. If $\mathbb{E}[e^{\lambda x}] < \infty$ for some $\lambda > 0$, then by Markov's inequality, we have

$$\mathbb{P}(X > t) = \mathbb{P}(e^{\lambda X} > e^{\lambda t}) \leq \frac{\mathbb{E}[e^{\lambda X}]}{e^{\lambda t}}. \tag{2.10}$$

Note that if the p-th moment of X exists, then the q-th moment of X also exists for all $0 < q < p$; this holds since if we write $|X|^p = |X|^p \mathbf{1}_{\{|X| \leq 1\}} + |X|^p \mathbf{1}_{\{|X| > 1\}}$, and note that $\mathbb{E}[|X|^p]$ is finite if and only if $\mathbb{E}[|X|^p \mathbf{1}_{\{|X| > 1\}}]$ is finite, then

$$\mathbb{E}[|X|^q] = \mathbb{E}[|X|^q \mathbf{1}_{\{|X| \leq 1\}}] + \mathbb{E}[|X|^q \mathbf{1}_{\{|X| > 1\}}] \leq 1 + \mathbb{E}[|X|^p \mathbf{1}_{\{|X| > 1\}}] < \infty. \tag{2.11}$$

2.3 Convergence of Random Variables

September 22nd.

Throughout, we shall assume that $(X_n)_{n \geq 1}$ is a sequence of random variables on $(\Omega, \mathcal{F}, \mathbb{P})$, and X is another random variable on the same probability space. We shall discuss various notions of convergence of the sequence (X_n) to X.

Definition 2.6. X_n converges to X as *almost sure convergence*, written $X_n \xrightarrow{\text{a.s.}} X$, if there exists a set $E \in \mathcal{F}$ with $\mathbb{P}(E) = 0$ such that

$$\lim_{n \to \infty} X_n(\omega) = X(\omega) \quad \text{for all } \omega \in \Omega \setminus E. \tag{2.12}$$

Definition 2.7. X_n converges to X as *convergence in probability*, written $X_n \xrightarrow{\text{P}} X$, if

$$\lim_{n\to\infty} \mathbb{P}(|X_n - X| > \epsilon) = 0 \quad \text{for all } \epsilon > 0. \tag{2.13}$$

Almost sure convergence is the stronger notion—

Proposition 2.8. *If $X_n \xrightarrow{\text{a.s.}} X$, then $X_n \xrightarrow{\text{P}} X$.*

Proof. Let $\epsilon > 0$. Since $X_n \xrightarrow{\text{a.s.}} X$, there exists a set $E \in \mathcal{F}$ with $\mathbb{P}(E) = 0$ such that $\lim_{n\to\infty} X_n(\omega) = X(\omega)$ for all $\omega \in \Omega\setminus E$. Therefore, for each $\omega \in \Omega\setminus E$, there exists $N_\epsilon(\omega) \in \mathbb{N}$ such that $|X_n(\omega) - X(\omega)| \le \epsilon$ for all $n \ge N_\epsilon(\omega)$. Now define

$$A_k := \{\omega \in \Omega \mid |X_n(\omega) - X(\omega)| \le \epsilon, \text{ for all } n \ge k\}. \tag{2.14}$$

Then $(A_k)_{k\ge1}$ is an increasing sequence of sets such that $\emptyset \subseteq A_k \uparrow \Omega \setminus E$; for any $(1 >)\delta > 0$, there exists a natural $N \in \mathbb{N}$ such that $\mathbb{P}(A_N) \ge 1 - \delta$. Then, we must have

$$\{|X_n - X| > \epsilon\} \subseteq \Omega \setminus A_N \implies 0 \le \mathbb{P}(|X_n - X| > \epsilon) \le \mathbb{P}(\Omega \setminus A_N) \le \delta. \tag{2.15}$$

Letting $\delta \to 0^+$, we get $\mathbb{P}(|X_n - X| > \epsilon) \to 0$ as $n \to \infty$; since $\epsilon > 0$ was arbitrary, we have $X_n \xrightarrow{\text{P}} X$. ■

However, the converse does not hold. Consider the sequence of random variables $(X_n)_{n\ge1}$ on $([0,1], \mathcal{B}([0,1]), \lambda)$ defined as

$$X_{2^k+j}(x) = \mathbf{1}_{[j2^{-k},(j+1)2^{-k})}(x), \qquad k = 0,1,2,\ldots, \quad j = 0,1,\ldots,2^k-1. \tag{2.16}$$

Let $X = 0$. For $n = 2^k + j$, we have

$$\mathbb{P}(|X_n - X| > \epsilon) = 2^{-k} \longrightarrow 0 \qquad (0 < \epsilon < 1), \tag{2.17}$$

and the probability is zero for $\epsilon \ge 1$. Hence $X_n \xrightarrow{\text{P}} 0$. On the other hand, every $x \in [0,1)$ belongs to exactly one interval at each level k, so $X_n(x) = 1$ infinitely often, while $X_n(x) = 0$ infinitely often. Thus X_n does not converge almost surely to 0 (indeed, it fails to converge for every $x \in [0,1)$).

Lemma 2.9. *Suppose $X_n \xrightarrow{\text{P}} X$ and, additionally,*

$$\sum_{n=1}^{\infty} \mathbb{P}(|X_n - X| > \epsilon) < \infty \qquad \textit{for all } \epsilon > 0. \tag{2.18}$$

Then $X_n \xrightarrow{\text{a.s.}} X$.

Proof. Let $\epsilon > 0$. By the Borel-Cantelli lemma, we have $\mathbb{P}(|X_n - X| > \epsilon \text{ i.o.}) = 0$. Let $E_\epsilon = \{|X_n - X| > \epsilon \text{ i.o.}\}$, so that $\mathbb{P}(E_\epsilon) = 0$. Then for all $\omega \in \Omega \setminus E_\epsilon$, there exists $N_\epsilon(\omega) \in \mathbb{N}$ such that $|X_n(\omega) - X(\omega)| \le \epsilon$ for all $n \ge N_\epsilon(\omega)$. Now define

$$E = \bigcup_{m=1}^{\infty} E_{1/m}. \tag{2.19}$$

Then $\mathbb{P}(E) = 0$, and for all $\omega \in \Omega \setminus E$, we have $|X_n(\omega) - X(\omega)| \le 1/m$ for all n sufficiently large, for every $m = 1, 2, \ldots$. Therefore, $\lim_{n\to\infty} X_n(\omega) = X(\omega)$ for all $\omega \in \Omega \setminus E$, and hence $X_n \xrightarrow{\text{a.s.}} X$. ■

Let $(X_n)_{n\ge1}$ be a sequence of pairwise independent random variables with $\mathbb{E}[X_n] = 0$ and $\text{Var}(X_n) \le M$ for all n and some constant $M > 0$. Then if $S_n = X_1 + \cdots + X_n$, we have $S_n/n \xrightarrow{\text{P}} 0$ since

$$\mathbb{P}(|S_n/n| > \epsilon) \le \frac{\mathbb{E}[|\frac{S_n}{n}|^2]}{\epsilon^2} = \frac{\text{Var}(S_n/n)}{\epsilon^2} = \frac{\text{Var}(S_n)}{n^2\epsilon^2} = \frac{\sum_{i=1}^n \text{Var}(X_i)}{n^2\epsilon^2} \le \frac{M}{n\epsilon^2} \to 0 \quad \text{as } n \to \infty. \tag{2.20}$$

Let $(X_n)_{n\geq 1}$ be bounded independent random variables with $\mathbb{E}[X_n] = 0$ for all n. Then if $S_n = X_1 + \cdots + X_n$, we have $S_n/n \xrightarrow{\text{a.s.}} 0$; to this end, we have

$$\mathbb{P}(|S_n/n| > \epsilon) \leq \frac{\mathbb{E}[|S_n/n|^4]}{\epsilon^4} = \frac{\mathbb{E}[S_n^4]}{n^4\epsilon^4} \leq \frac{C(n+n^2)M^4}{n^4\epsilon^4} \lesssim \frac{1}{n^2}. \tag{2.21}$$

Thus,

$$\sum_{n=1}^{\infty} \mathbb{P}(|S_n/n| > \epsilon) < \infty, \tag{2.22}$$

or $|S_n/n| \xrightarrow{\text{a.s.}} 0$ by the previous lemma.

2.4 Second Moment Method

September 25th.

Let $X \geq 0$ be a non-negative random variable. Given just this, note that $\mathbb{P}(X > 0)$ cannot have a lower bound since we could very well define X to be 0 with probability $1 - \epsilon$ and 1 with probability ϵ, for any $\epsilon > 0$. If we specify that $\mathbb{E}[X]$ has to be 'large', we stil cannot guarantee that $\mathbb{P}(X > 0)$ is large, since we could define X to be 0 with probability $1 - \epsilon$ and $1/\epsilon$ with probability ϵ, for any $1 > \epsilon > 0$.

Suppose we are give $\mu = \mathbb{E}[X]$. Then by Markov's inequality, we have

$$\mathbb{P}(X > \frac{\mu}{2}) = \mathbb{P}(|X - \mu| \leq \frac{\mu}{2}) + \mathbb{P}(X > \frac{3\mu}{2}) \geq 1 - \mathbb{P}(|X - \mu| > \frac{\mu}{2}) \geq 1 - \frac{4\operatorname{Var}(X)}{\mu^2} \geq 1 - 4\delta \tag{2.23}$$

when $\operatorname{Var}(X) \leq \delta\mu^2$ for some $\delta > 0$. Thus, if $\operatorname{Var}(X)$ is small compared to $\mathbb{E}[X]^2$, then X is concentrated around its mean, and hence $\mathbb{P}(X > 0)$ is large. Note that the choice of $\mu/2$ is arbitrary; we have the following generalization.

Lemma 2.10 (The *second moment method*)**.** *Let $X \geq 0$ be a non-negative random variable, and $0 \leq \alpha \leq 1$. Then*

$$\mathbb{P}(X > \alpha\mathbb{E}[X]) \geq (1-\alpha)^2 \frac{\mathbb{E}[X]^2}{\mathbb{E}[X^2]}. \tag{2.24}$$

In particular,

$$\mathbb{P}(X > 0) \geq \frac{\mathbb{E}[X]^2}{\mathbb{E}[X^2]}. \tag{2.25}$$

Proof. Recall the Cauchy-Schwarz inequality, which states that for any random variables X and Y, we have $\mathbb{E}[XY]^2 \leq \mathbb{E}[X^2]\mathbb{E}[Y^2]$. Let $Y = \mathbf{1}_{\{X > \alpha\mathbb{E}[X]\}}$. Then $\mathbb{E}[Y^2] = \mathbb{E}[Y] = \mathbb{P}(X > \alpha\mathbb{E}[X])$, and $\mathbb{E}[XY] = \mathbb{E}[X\mathbf{1}_{\{X>\alpha\mathbb{E}[X]\}}] = \mathbb{E}[X] - \mathbb{E}[X\mathbf{1}_{\{X\leq\alpha\mathbb{E}[X]\}}] \geq (1-\alpha)\mathbb{E}[X]$. Therefore, by Cauchy-Schwarz,

$$\mathbb{P}(X > \alpha\mathbb{E}[X]) = \mathbb{E}[Y^2] \geq \frac{\mathbb{E}[XY]^2}{\mathbb{E}[X^2]} \geq (1-\alpha)^2 \frac{\mathbb{E}[X]^2}{\mathbb{E}[X^2]}. \tag{2.26}$$

■

We may also prove the Borel-Cantelli lemma using the second moment method, the statement being that if $A_1, A_2, \ldots$ are independent events (or simply pairwise independent) with $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$, then $\mathbb{P}(A_n \text{ i.o.}) = 1$.

Proof. Let $B_n = \bigcup_{i=n}^{\infty} A_i$. Then $B_n \downarrow \{A_n \text{ i.o.}\}$. Fix n and define $S_k := \sum_{j=n}^{k} \mathbf{1}_{A_j}$ for $k \geq n$. Then, with

$\mathbb{P}(A_i) = \alpha_i$,

$$\begin{aligned}\mathbb{P}(S_k \geq 1) = \mathbb{P}(S_k > 0) \geq \frac{\mathbb{E}[S_k]^2}{\mathbb{E}[S_k^2]} &= \frac{(\sum_{j=n}^k \mathbb{P}(A_j))^2}{\sum_{j=n}^k \mathbb{P}(A_j) + 2\sum_{n\leq i<j\leq k} \mathbb{P}(A_i \cap A_j)} \\ &= \frac{(\sum_{j=n}^k \alpha_j)^2}{\sum_{j=n}^k \alpha_j + 2\sum_{n\leq i<j\leq k} \alpha_i\alpha_j} = \frac{(\sum_{i=n}^k \alpha_i)^2}{(\sum_{i=n}^k \alpha_i) + (\sum_{i=n}^k \alpha_i)^2} \\ &\geq \frac{1}{1 + \frac{1}{\sum_{i=n}^k \alpha_i}}. \end{aligned} \tag{2.27}$$

Letting $k \to \infty$, we get

$$\mathbb{P}(B_n) = \lim_{k\to\infty} \mathbb{P}(S_k \geq 1) \geq 1. \tag{2.28}$$

Since this holds for all n, we have $\mathbb{P}(A_n \text{ i.o.}) = 1$. ■

2.4.1 The Coupon Collector Problem

n different books are placed on a bookshelf. Every night, a book is chosen uniformly at random, read, and then returned to the shelf. Let T_n be the number of days until all n books have been read. We want to find $\mathbb{E}[T_n]$. To this end, define $Z_1 = 1$, and for $2 \leq k \leq n$, let Z_k be the number of days after the first $k-1$ books have been read until a new book is read. Then $T_n = Z_1 + Z_2 + \cdots + Z_n$. Note that $Z_2 \sim \text{Geom}(\frac{n-1}{n})$, since there are $n-1$ books that have not been read yet with probability $\frac{n-1}{n}$ of reading a new book, and similarly $Z_k \sim \text{Geom}(\frac{n-k+1}{n})$ for $2 \leq k \leq n$. Therefore, we have

$$\mathbb{E}[T_n] = \sum_{k=1}^n \mathbb{E}[Z_k] = \sum_{k=1}^n \frac{n}{n-k+1} = n\sum_{k=1}^n \frac{1}{k} \sim n\log n. \tag{2.29}$$

Now let θ_n be a sequence of positive numbers such that $\theta_n \to \infty$ as $n \to \infty$. We claim that

$$\mathbb{P}(|T_n - n\log n| \geq n\theta_n) \to 0 \quad \text{as } n \to \infty. \tag{2.30}$$

To show this, set $t \in \mathbb{N}$, and define

$$S_t := \sum_{j=1}^n \mathbf{1}_{B_j}, \tag{2.31}$$

where B_j is the event that book j has not been picked in the first t days. Then S_t is the number of books that have not been picked in the first t days, and hence $\{T_n > t\} = \{S_t \geq 1\} = \{S_t > 0\}$. Computing moments of S_t, while later setting $t = n\log n + n\theta_n$, we have

$$\mathbb{E}[S_t] = \sum_{j=1}^n \mathbb{P}(B_j) = n\left(1 - \frac{1}{n}\right)^t \geq ne^{-(\frac{1}{n} + \frac{1}{n^2})t} = e^{\theta_n - \frac{1}{n}(\log n - \theta_n)}. \tag{2.32}$$

and

$$\begin{aligned}\mathbb{E}[S_t^2] = \sum_{j=1}^n \mathbb{P}(B_j) + 2\sum_{1\leq i<j\leq n} \mathbb{P}(B_i \cap B_j) &= n\left(1 - \frac{1}{n}\right)^t + n(n-1)\left(1 - \frac{2}{n}\right)^t \\ &\leq ne^{-\frac{t}{n}} + n^2 e^{-\frac{2t}{n}} = e^{\theta_n} + e^{2\theta_n}. \end{aligned} \tag{2.33}$$

Therefore, by the second moment method, we have

$$\mathbb{P}(T_n > n\log n - n\theta_n) = \mathbb{P}(S_t > 0) \geq \frac{\mathbb{E}[S_t]^2}{\mathbb{E}[S_t^2]} \geq \frac{e^{2\theta_n - \frac{2}{n}(\log n - \theta_n)}}{e^{\theta_n} + e^{2\theta_n}} \implies \mathbb{P}(T_n > n\log n - n\theta_n) \xrightarrow{n\to\infty} 1. \tag{2.34}$$

On the other hand, setting $t = n\log n + n\theta_n$ and using Markov's inequality, we have

$$\begin{aligned}\mathbb{P}(T_n > n\log n + n\theta_n) = \mathbb{P}(S_t \geq 1) \leq \mathbb{E}[S_t] \leq n\left(1 - \frac{1}{n}\right)^t \leq ne^{-\frac{1}{n}n(\log n + \theta_n)} = e^{-\theta_n} \\ \implies \mathbb{P}(T_n > n\log n + n\theta_n) \xrightarrow{n\to\infty} 0. \end{aligned} \tag{2.35}$$

Therefore, we conclude that

$$\mathbb{P}(|T_n - n\log n| \geq n\theta_n) = \mathbb{P}(T_n > n\log n + n\theta_n) + \mathbb{P}(T_n < n\log n - n\theta_n) \xrightarrow{n\to\infty} 0. \tag{2.36}$$

Dividing by $n\log n$, and setting $\theta_n = \epsilon \log n$ for any $\epsilon > 0$, we get

$$\frac{T_n}{n\log n} \xrightarrow{\mathrm{P}} 1. \tag{2.37}$$

2.4.2 Random Series

Let $(a_n)_{n\geq 0}$ be a sequence of independent and identically distribution random variables with $a_n \sim \mathrm{Exp}(1)$, which has a density function $e^{-x}\mathbf{1}_{x\geq 0}$. Now define

$$f(t) = \sum_{n=0}^{\infty} a_n t^n \qquad \text{for all } t \in \mathbb{R}. \tag{2.38}$$

This is a random power series, and we want to find the radius of convergence, R, of this series. We know that R is related to the coefficients a_n as

$$\frac{1}{R} = \limsup_{n\to\infty} |a_n|^{1/n}. \tag{2.39}$$

This limit is a random variable which is measurable with respect to the tail-σ-algebra of the sequence $(a_n)_{n\geq 0}$, and hence by Kolmogorov's zero-one law, there exists a constant $c \in [0,\infty]$ such that $\mathbb{P}(R = c) = 1$. We claim that $c = 1$. First note that if we set $t = 1+\epsilon$ for some $\epsilon > 0$, then

$$\mathbb{P}(|a_n| > t^n) = e^{-t^n} \leq \frac{1}{(1+\epsilon)^n} \quad \text{for all } n \text{ sufficiently large.} \tag{2.40}$$

Thus, $\sum_{n=0}^{\infty} \mathbb{P}(|a_n| > t^n) < \infty$, and hence by the Borel-Cantelli lemma, we have $\mathbb{P}(|a_n|^{1/n} > t \text{ i.o.}) = 0$, or $\mathbb{P}(\limsup_{n\to\infty} |a_n|^{1/n} > t) = 0$. Since this holds for all $t > 1$, we have $\mathbb{P}(\limsup_{n\to\infty} |a_n|^{1/n} > 1) = 0$. On the other hand, if we set $t = 1-\epsilon$ for some $\epsilon > 0$, then

$$\sum_{n=0}^{\infty} \mathbb{P}(|a_n| > t^n) = \sum_{n=0}^{\infty} e^{-t^n} = \infty, \tag{2.41}$$

so that by the Borel-Cantelli lemma, we have $\mathbb{P}(|a_n|^{1/n} > t \text{ i.o.}) = 1$, or $\mathbb{P}(\limsup_{n\to\infty} |a_n|^{1/n} > t) = 1$. Since this holds for all $t < 1$, we have $\mathbb{P}(\limsup_{n\to\infty} |a_n|^{1/n} < 1) = 0$. Therefore, we conclude that $\mathbb{P}(\limsup_{n\to\infty} |a_n|^{1/n} = 1) = 1$, and hence $\mathbb{P}(R = 1) = 1$.

2.5 Law of Large Numbers

September 29th.

Theorem 2.11 (The *basic weak law of large numbers*)**.** *Let $(X_n)_{n\geq 1}$ be a sequence of independent and identically distributed random variables with $\mathbb{E}[|X_1|] < \infty$ and $\mathrm{Var}(X_1) < \infty$. Then*

$$\frac{S_n}{n} = \frac{X_1 + \cdots + X_n}{n} \xrightarrow{\mathrm{P}} \mu = \mathbb{E}[X_1]. \tag{2.42}$$

Proof. We wish to show that for all $\epsilon > 0$, we have $\mathbb{P}(|S_n/n - \mu| > \epsilon) \to 0$ as $n \to \infty$. To this end, we first note that by Chebyshev's inequality, we have

$$\mathbb{P}\left(\left|\frac{S_n}{n} - \mu\right| > \epsilon\right) = \mathbb{P}\left(\left|\sum_{i=1}^{n}(X_i - \mu)\right| > n\epsilon\right) \leq \frac{\mathbb{E}\left[\left(\sum_{i=1}^{n}(X_i-\mu)\right)^2\right]}{n^2\epsilon^2} = \frac{n\,\mathrm{Var}(X_1)}{n^2\epsilon^2} = \frac{\mathrm{Var}(X_1)}{n\epsilon^2} \xrightarrow{n\to\infty} 0. \tag{2.43}$$

■

We can use the weak law of large numbers to prove the Weierstrass approximation theorem, which states that any continuous function on $[0,1]$ can be uniformly approximated by polynomials. To this end, we make use of the *Bernstein polynomials*, which are defined as follows. For a continuous function $f : [0,1] \to \mathbb{R}$, define the n-th Bernstein polynomial of f as

$$Q_{n,f} : [0,1] \to \mathbb{R}, \qquad Q_{n,f}(p) = \sum_{k=0}^{n} f\left(\frac{k}{n}\right)\binom{n}{k} p^k (1-p)^{n-k}. \tag{2.44}$$

Note that if $S_n \sim \text{Bin}(n,p)$, then $\mathbb{P}(S_n = k) = \binom{n}{k} p^k (1-p)^{n-k}$, and hence $Q_{n,f}(p) = \mathbb{E}[f(S_n/n)]$. Let $S_n = X_1 + X_2 + \cdots + X_n$, where $X_1, \ldots, X_n \sim \text{Ber}(p)$ are independent. Then, we have

$$|Q_{n,f}(p) - f(p)| \le \mathbb{E}\left[\left|f\left(\frac{S_n}{n}\right) - f(p)\right|\right]. \tag{2.45}$$

Splitting via the indicator function of $\{\left|\frac{S_n}{n} - p\right| > \delta\}$ for some $\delta > 0$ we choose later, we have

$$\mathbb{E}\left[\left|f\left(\frac{S_n}{n}\right) - f(p)\right|\right] = \mathbb{E}\left[\left|f\left(\frac{S_n}{n}\right) - f(p)\right| \cdot \mathbf{1}_{|\frac{S_n}{n} - p| \le \delta}\right] + \mathbb{E}\left[\left|f\left(\frac{S_n}{n}\right) - f(p)\right| \cdot \mathbf{1}_{|\frac{S_n}{n} - p| > \delta}\right]. \tag{2.46}$$

By the uniform continuity of f, given $\epsilon > 0$, there exists $\delta > 0$ such that $|f(x) - f(y)| < \epsilon/2$ whenever $|x - y| \le \delta$. Thus, the first of these terms may simply be bounded by $\epsilon/2$. For the second term, we have

$$\mathbb{E}\left[\left|f\left(\frac{S_n}{n}\right) - f(p)\right| \cdot \mathbf{1}_{|\frac{S_n}{n} - p| > \delta}\right] \le 2\|f\|_\infty \mathbb{P}\left(\left|\frac{S_n}{n} - p\right| > \delta\right) \le 2\|f\|_\infty \frac{\text{Var}(S_n/n)}{\delta^2} = 2\|f\|_\infty \frac{p(1-p)}{n\delta^2}. \tag{2.47}$$

With $c_f = 2\|f\|_\infty (1/4)$, the second term is bounded by $c_f/(n\delta^2)$, which goes to 0 as $n \to \infty$. Therefore, we have $|Q_{n,f}(p) - f(p)| < \epsilon$ for all n sufficiently large, and hence $Q_{n,f} \to f$ uniformly on $[0,1]$.

We now prove the weak law of large numbers, where we drop the assumption of finite variance.

Theorem 2.12 (The *weak law of large numbers*)**.** *Let $(X_n)_{n\ge1}$ be a sequence of independent and identically distributed random variables with $\mathbb{E}[|X_1|] < \infty$. Then*

$$\frac{S_n}{n} = \frac{X_1 + \cdots + X_n}{n} \xrightarrow{\text{P}} \mu = \mathbb{E}[X_1]. \tag{2.48}$$

Proof. Note that it suffices to prove the result for $\mu = 0$, since we can always replace X_n with $X_n - \mu$. Since we have the basic weak law of large numbers for random variables with finite variance, we will approximate X_n with a truncated version of itself to reduce to the finite variance case. To this end, for each $n \ge 1$, we can split X_j into two parts as

$$X_j = X_j \mathbf{1}_{|X_j| \le A_n} + X_j \mathbf{1}_{|X_j| > A_n} = Y_{j,n} + Z_{j,n}, \tag{2.49}$$

where $(A_n)_{n\ge1}$ is a sequence of positive numbers the diverge properly to infinity, that we choose later. The first of these terms converges pointwise to X_j as $n \to \infty$, and is bounded uniformly; by the dominated convergence theorem, we have $\mathbb{E}[X_j \mathbf{1}_{|X_j| \le A_n}] \to \mathbb{E}[X_j] = 0$ as $n \to \infty$. The second of these terms converges pointwise to 0 as $n \to \infty$, and is bounded by $|X_j|$; again by the dominated convergence theorem, we have $\mathbb{E}[X_j \mathbf{1}_{|X_j| > A_n}] \to 0$ as $n \to \infty$. Let $\mu_n = \mathbb{E}[Y_{1,n}] = -\mathbb{E}[Z_{1,n}]$. Then $\mu_n \to 0$ as $n \to \infty$. Now, each S_n may be split as $S_n = S_n^Y + S_n^Z$, where $S_n^Y = Y_{1,n} + \cdots + Y_{n,n}$ and $S_n^Z = Z_{1,n} + \cdots + Z_{n,n}$. Then, for any $\epsilon > 0$, we have

$$\mathbb{P}\left(\left|\frac{S_n}{n}\right| > \delta\right) = \mathbb{P}\left(\left|\frac{S_n^Y}{n} + \frac{S_n^Z}{n}\right| > \delta\right) \le \mathbb{P}\left(\left|\frac{S_n^Y}{n} - \mu_n\right| > \frac{\delta}{2}\right) + \mathbb{P}\left(\left|\frac{S_n^Z}{n} + \mu_n\right| > \frac{\delta}{2}\right). \tag{2.50}$$

Via Chebyshev's, the first of these terms is bounded as

$$\mathbb{P}\left(\left|\frac{S_n^Y}{n} - \mu_n\right| > \frac{\delta}{2}\right) \le \frac{4\,\text{Var}(Y_{1,n})}{n\delta^2} \tag{2.51}$$

For $\delta > 0$ fixed, let us estimate this for large enough n such that $|\mu_n| < \frac{\delta}{4}$. For the second term, we have

$$\mathbb{P}\left(\left|\frac{S_n^Z}{n} + \mu_n\right| > \frac{\delta}{2}\right) \leq \mathbb{P}\left(\left|\frac{S_n^Z}{n}\right| > \frac{\delta}{4}\right) \leq \mathbb{P}(\text{at least one of the } Z_{j,n} \text{ is non-zero}) \leq n\mathbb{P}(|X_1| > A_n). \tag{2.52}$$

Combining, we obtain

$$\mathbb{P}\left(\left|\frac{S_n}{n}\right| > \delta\right) \leq \frac{4\operatorname{Var}(Y_{1,n})}{n\delta^2} + n\mathbb{P}(|X_1| > A_n) \leq \frac{4A_n\mathbb{E}[|X_1|]}{n\delta^2} + \frac{n\mathbb{E}[|X_1| \cdot \mathbf{1}_{|X_1|>A_n}]}{A_n}. \tag{2.53}$$

We now wish to choose A_n such that both of these terms go to 0 as $n \to \infty$. So, given $\epsilon > 0$, choosen $A_n = \epsilon n\delta^2/4\mathbb{E}[|X_1|]$. Then, we have

$$\mathbb{P}\left(\left|\frac{S_n}{n}\right| > \delta\right) \leq \epsilon + \frac{4\mathbb{E}[|X_1|]}{\delta^2}\epsilon\mathbb{E}[|X_1| \cdot \mathbf{1}_{|X_1|>A_n}] \xrightarrow{n\to\infty} 2\epsilon. \tag{2.54}$$

Since $\epsilon > 0$ was arbitrary, we conclude that $\mathbb{P}(|S_n/n| > \delta) \to 0$ as $n \to \infty$, and hence $S_n/n \xrightarrow{\mathrm{P}} 0$. ■

Example 2.13. Let $X_1, X_2, \ldots \sim \mathrm{Exp}(1)$ be independent and identically distributed random variables, and set $M_n := \max\{X_1, \ldots, X_n\}$. Then $M_n/\log n \xrightarrow{\mathrm{P}} 1$ as $n \to \infty$. To see this, note that

$$\mathbb{P}(M_n > (1+\epsilon)\log n) = 1 - \mathbb{P}(X_1 \leq (1+\epsilon)\log n)^n = 1 - (1 - n^{-(1+\epsilon)})^n \to 0, \tag{2.55}$$

as $n \to \infty$.

2.6 L^p Spaces

October 6th.

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. For $0 < p < \infty$, define the set

$$V_p = \{X : \Omega \to \mathbb{R} \mid \mathbb{E}[|X|^p] < \infty\} \tag{2.56}$$

and (for $p = \infty$) define

$$V_\infty = \{X : \Omega \to \mathbb{R} \mid \exists M > 0 \text{ such that } \mathbb{P}(|X| > M) = 0\}. \tag{2.57}$$

Proposition 2.14. *V_p and V_∞ are real vector spaces with respect to the obvious operations of addition and scalar multiplication.*

Proof. For $p \geq 1$, let $X, Y \in V_p$ and $\alpha \in \mathbb{R}$. Then, by the triangle inequality and convexity of $x \mapsto x^p$, we have

$$\mathbb{E}[|X+Y|^p] \leq 2^{p-1}(\mathbb{E}[|X|^p] + \mathbb{E}[|Y|^p]) < \infty, \tag{2.58}$$

and

$$\mathbb{E}[|\alpha X|^p] = |\alpha|^p\,\mathbb{E}[|X|^p] < \infty. \tag{2.59}$$

Hence V_p is closed under addition and scalar multiplication. For $p < 1$, we use the fact that $|x+y|^p \leq |x|^p + |y|^p$, and the fact that $|\alpha x|^p = |\alpha|^p\,|x|^p$. The proof for V_∞ is similar. ■

We now wish to define a norm on V_p and V_∞; the obvious candidate is to define $\|X\|_p = \mathbb{E}[|X|^p]^{1/p}$ for $0 < p < \infty$, and $\|X\|_\infty = \inf\{M > 0 \mid \mathbb{P}(|X| > M) = 0\}$. However, this is not a norm since $\|X\|_p = 0$ implies $X = 0$ only almost surely and not exactly. The definiteness property fails. Hence, to rectify this, we define an equivalence relation on V_p and V_∞ as follows. For $X, Y \in V_p$, we say that $X \sim Y$ if $\mathbb{P}(X \neq Y) = 0$, and similarly for $X, Y \in V_\infty$. Then, we define the quotient spaces

$$L^p = V_p/\sim \qquad \text{and} \qquad L^\infty = V_\infty/\sim. \tag{2.60}$$

The norms $\|\cdot\|_p$ and $\|\cdot\|_\infty$ are now definite on L^p and L^∞, respectively. For $p < 1$, the triangle inequality still fails. Hence, L_∞ and L_p for $p \geq 1$ are normed vector spaces. In fact, these are Banach spaces via the Riesz-Fischer theorem, which states that L^p and L^∞ are complete with respect to their respective norms. We will assume this theorem without proof. The L^2 space is particularly important, since it is a Hilbert space with respect to the inner product $\langle X, Y\rangle = \mathbb{E}[XY]$.

Proposition 2.15. *Let $1 \leq p \leq q \leq \infty$. Then $L^q \subseteq L^p$.*

Proof. If $q = \infty$, then $|X| \leq \|X\|_\infty$ almost surely, and hence $X \in L^p$. Now suppose that $q < \infty$ and let $X \in L^q$. If $p = q$, the result is immediate. For $p < q$, we have

$$\mathbb{E}[|X|^p] = \mathbb{E}[|X|^p \mathbf{1}_{|X|\leq 1}] + \mathbb{E}[|X|^p \mathbf{1}_{|X|>1}] \leq 1 + \mathbb{E}[|X|^q \mathbf{1}_{|X|>1}] \leq 1 + \mathbb{E}[|X|^q] < \infty, \tag{2.61}$$

and hence $X \in L^p$. ■

To show triangle inequality for L^p, $p \geq 1$, we require convex functions.

Definition 2.16. A function $\varphi : (a, b) \to \mathbb{R}$ is a *convex function* if φ is continuous and given any $u \in (a, b)$, there exists a linear function $L_u(x) = \alpha_u x + \beta_u$ such that $L_u(u) = \varphi(u)$ and $L_u(x) \leq \varphi(x)$ for all $x \in (a, b)$. In other words, there exists a line passing through $(u, \varphi(u))$ that lies below the graph of φ.

Lemma 2.17 (*Jensen's inequality*)**.** *Let $\varphi : \mathbb{R} \to \mathbb{R}$ be a convex function, and suppose $X, \varphi(X) \in L^1$. Then $\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)]$.*

Proof. Set $a = \mathbb{E}[X]$. Then, by the definition of convexity, there exists a linear function $y = m(x-a)+\varphi(a)$ such that $y \leq \varphi(x)$ for all $x \in \mathbb{R}$, or $m(X - a) + \varphi(a) \leq \varphi(X)$. Taking expectations gives

$$\varphi(\mathbb{E}[X]) = \varphi(a) \leq \mathbb{E}[\varphi(X)]. \tag{2.62}$$

■

Lemma 2.18 (*Minkowski's inequality*)**.** *Let $1 \leq p < \infty$. Then for any $X, Y \in L^p$, we have*

$$\|X + Y\|_p \leq \|X\|_p + \|Y\|_p . \tag{2.63}$$

.

Proof. Set $X' = X/\|X\|_p$ and $Y' = Y/\|Y\|_p$. Then $\|X'\|_p = \|Y'\|_p = 1$. Also set $a = \frac{\|X\|_p}{\|X\|_p+\|Y\|_p}$ and $b = \frac{\|Y\|_p}{\|X\|_p+\|Y\|_p}$. Since $x \mapsto x^p$ is convex, we have

$$|aX' + bY'|^p \leq a\,|X'|^p + b\,|Y'|^p \implies \mathbb{E}[|aX' + bY'|^p] \leq a\mathbb{E}[|X'|^p] + b\mathbb{E}[|Y'|^p] = a + b = 1. \tag{2.64}$$

Rearranging gives

$$\begin{aligned} \|X + Y\|_p &= (\mathbb{E}[|X + Y|^p])^{1/p} = (\mathbb{E}[|(a + b)(X' + Y')|^p])^{1/p} \\ &= (a + b)(\mathbb{E}[|aX' + bY'|^p])^{1/p} \leq a + b = \|X\|_p + \|Y\|_p . \end{aligned} \tag{2.65}$$

■

Theorem 2.19 (*Hölder's inequality*)**.** *Let $1 < p, q < \infty$ be such that $\frac{1}{p} + \frac{1}{q} = 1$. Then for any $X \in L^p$ and $Y \in L^q$, we have $XY \in L^1$ and*

$$\mathbb{E}[|XY|] \leq \|X\|_p \|Y\|_q . \tag{2.66}$$

Note that the case $p = q = 2$ is a special case of this theorem, and is known as the Cauchy-Schwarz inequality.

Proof. We first show that $\frac{a^p}{p} + \frac{b^q}{q} \geq ab$ for all $a, b > 0$. To see this, note that the function $f(a) = \frac{a^p}{p} - ab + \frac{b^q}{q}$ is convex in a, and has a unique minimum at $a = b^{1/(p-1)}$. Evaluating f at this point gives

$$f(b^{1/(p-1)}) = \frac{b^{p/(p-1)}}{p} - b^{p/(p-1)} + \frac{b^q}{q} = b^q \left(\frac{1}{p} - 1 + \frac{1}{q}\right) = 0. \tag{2.67}$$

This shows that $\frac{a^p}{p} + \frac{b^q}{q} \geq ab$ for all $a, b > 0$. Now, let $X \in L^p$ and $Y \in L^q$, and set $X' = X/\|X\|_p$ and $Y' = Y/\|Y\|_q$. Then $\|X'\|_p = \|Y'\|_q = 1$. Applying the inequality above with $a = |X'|$ and $b = |Y'|$, we have

$$|X'Y'| \leq \frac{|X'|^p}{p} + \frac{|Y'|^q}{q} \implies \mathbb{E}[|X'Y'|] \leq \frac{\mathbb{E}[|X'|^p]}{p} + \frac{\mathbb{E}[|Y'|^q]}{q} = \frac{1}{p} + \frac{1}{q} = 1. \tag{2.68}$$

Multiplying both sides by $\|X\|_p \|Y\|_q$ gives the desired result. ■

With the norm defined on L^p, we can define convergence in L^p by saying that a sequence $(X_n)_{n\geq 1}$ converges to X in L^p if $\|X_n - X\|_p \to 0$ as $n \to \infty$.

2.7 Central Limit Theorem

October 9th.

Recall that central limit theorem.

Theorem 2.20 (The *central limit theorem*)**.** *Let $X_1, X_2, \ldots$ be a sequence of independent and identically distributed random variables with $\mathbb{E}[X_1] = 0$ and $\mathrm{Var}(X) = 1$. Then*

$$\frac{S_n}{\sqrt{n}} = \frac{X_1 + \cdots + X_n}{\sqrt{n}} \xrightarrow{d} X \tag{2.69}$$

where $X \sim N(0, 1)$.

The proof of this theorem will be proved in full generality later; for now, we make some further assumptions to simplify the proof.

Lemma 2.21. *Let $(Y_n)_{n\geq 1}$ be a sequence of random variables, and Y a random variable. Then $Y_n \xrightarrow{d} Y$ if and only if $\mathbb{E}[f(Y_n)] \to \mathbb{E}[f(Y)]$ for all bounded continuous functions $f : \mathbb{R} \to \mathbb{R}$.*

Proof. Suppose first that $Y_n \xrightarrow{d} Y$. Then, by the definition of convergence in distribution, $\mathbb{E}[f(Y_n)] \to \mathbb{E}[f(Y)]$ for every bounded continuous function $f : \mathbb{R} \to \mathbb{R}$.

Conversely, suppose that $\mathbb{E}[f(Y_n)] \to \mathbb{E}[f(Y)]$ for all bounded continuous functions $f : \mathbb{R} \to \mathbb{R}$. Let F_n and F be the cumulative distribution functions of Y_n and Y, respectively. Fix a continuity point t of F. For each $k \geq 1$, define the function $f_k : \mathbb{R} \to \mathbb{R}$ as

$$f_k(x) = \begin{cases} 1 & \text{if } x \leq t, \\ 1 - k(x - t) & \text{if } t < x \leq t + \frac{1}{k}, \\ 0 & \text{if } x > t + \frac{1}{k}. \end{cases} \tag{2.70}$$

Note that for any random variable X,

$$F_X(t) \leq \mathbb{E}[f_k(X)] \leq F_X(t + 1/k). \tag{2.71}$$

Therefore, for fixed k,

$$\limsup_{n\to\infty} F_n(t) \leq \mathbb{E}[f_k(Y)] \leq F(t + 1/k). \tag{2.72}$$

Letting $k \to \infty$ and using the continuity of F at t gives

$$\limsup_{n\to\infty} F_n(t) \leq F(t). \tag{2.73}$$

To obtain the reverse inequality, define the bounded continuous function $g_k : \mathbb{R} \to \mathbb{R}$ by

$$g_k(x) = \begin{cases} 0 & \text{if } x < t - \frac{1}{k}, \\ k(x - t) + 1 & \text{if } t - \frac{1}{k} \leq x < t, \\ 1 & \text{if } x \geq t. \end{cases} \tag{2.74}$$

Since $\mathbf{1}_{[t,\infty)} \le g_k$, we have

$$1 - F_n(t) = \mathbb{P}(Y_n > t) \le \mathbb{E}[g_k(Y_n)] \to \mathbb{E}[g_k(Y)] \le \mathbb{P}(Y \ge t - 1/k) = 1 - F(t - 1/k). \tag{2.75}$$

Hence

$$\liminf_{n\to\infty} F_n(t) \ge F(t - 1/k). \tag{2.76}$$

Letting $k \to \infty$ and using the continuity of F at t gives $\liminf_{n\to\infty} F_n(t) \ge F(t)$. Thus $F_n(t) \to F(t)$ at every continuity point of F, and consequently $Y_n \xrightarrow{d} Y$. ■

2.7.1 Characteristic Functions

For a random variable X, we define the *characteristic function* of X as

$$\psi_X : \mathbb{R} \to \mathbb{C}, \qquad \psi_X(t) = \mathbb{E}[e^{itX}] = \mathbb{E}[\cos(tX)] + i\mathbb{E}[\sin(tX)]. \tag{2.77}$$

Lemma 2.22. *Let $(Y_n)_{n\ge1}$ be a sequence of random variables, and Y a random variable. Then $\mathbb{E}[e^{itY_n}] \to \mathbb{E}[e^{itY}]$ for all $t \in \mathbb{R}$ implies that $Y_n \xrightarrow{d} Y$.*

Proof. Let ψ_n and ψ denote the characteristic functions of Y_n and Y, respectively. We first show that $(Y_n)_{n\ge1}$ is tight. Fix $T > 0$. By the assumed pointwise convergence and dominated convergence,

$$\sup_{n\ge1} \int_{-T}^{T} (1 - \Re\psi_n(t))\, dt < \infty. \tag{2.78}$$

On the other hand,

$$\int_{-T}^{T} (1 - \cos(tx))\, dt = 2T - \frac{2\sin(Tx)}{x} \ge 2T - \frac{2}{|x|}. \tag{2.79}$$

Thus, for $R > 0$ with $2T - 2/R > 0$,

$$\sup_{n\ge1} \mathbb{P}(|Y_n| > R) \le \frac{1}{2T - 2/R} \sup_{n\ge1} \int_{-T}^{T} (1 - \Re\psi_n(t))\, dt. \tag{2.80}$$

First choosing T sufficiently large and then R sufficiently large shows that $(Y_n)_{n\ge1}$ is tight.

By tightness, every subsequence of $(Y_n)_{n\ge1}$ has a further subsequence, denoted by $(Y_{n_j})_{j\ge1}$, which converges in distribution to some random variable W. For each $t \in \mathbb{R}$, the first lemma gives

$$\mathbb{E}[e^{itW}] = \lim_{j\to\infty} \mathbb{E}[e^{itY_{n_j}}] = \mathbb{E}[e^{itY}]. \tag{2.81}$$

Therefore W and Y have the same distribution, since characteristic functions determine distributions. Hence every subsequence has a further subsequence converging in distribution to Y, which implies that $Y_n \xrightarrow{d} Y$. ■

Using this lemma, we may prove the central limit theorem assuming finite third moment.

Proof of the central limit theorem. By the above lemma, it suffices to show that $\mathbb{E}[e^{itS_n/\sqrt{n}}] \to \psi_Z(t)$ for all $t \in \mathbb{R}$, where $Z \sim N(0,1)$. Via independence, we have

$$\mathbb{E}\left[e^{itS_n/\sqrt{n}}\right] = \psi_{X_1}\left(\frac{t}{\sqrt{n}}\right)^n. \tag{2.82}$$

Since $\mathbb{E}[X_1] = 0$ and $\mathbb{E}[X_1^2] = 1$, Taylor's theorem gives

$$\psi_{X_1}(u) = 1 - \frac{u^2}{2} + o(u^2) \qquad (u \to 0). \tag{2.83}$$

Consequently,

$$\psi_{X_1}\left(\frac{t}{\sqrt{n}}\right)^n = \left(1 - \frac{t^2}{2n} + o\left(\frac{1}{n}\right)\right)^n \to e^{-t^2/2}. \tag{2.84}$$

Finally, if $Z \sim N(0,1)$, then completing the square in the Gaussian integral gives

$$\psi_Z(t) = \mathbb{E}[e^{itZ}] = e^{-t^2/2}. \tag{2.85}$$

Therefore $\mathbb{E}[e^{itS_n/\sqrt{n}}] \to \psi_Z(t)$ for every $t \in \mathbb{R}$, and the above lemma proves the result. ■

October 13th.

We now prove the central limit theorem in full generality.

Proof of the central limit theorem. Let $X_1, X_2, \dots$ be independent and identically distributed random variables with $\mathbb{E}[X_1] = 0$ and $\mathrm{Var}(X_1) = 1$. Now let $Y_1, Y_2, \dots$ be independent and identically distributed random variables with $Y_1 \sim N(0,1)$. Now, for $0 \le k \le n$, define

$$V_k := \frac{1}{\sqrt{n}}(X_1 + X_2 + \cdots + X_k + Y_{k+1} + Y_{k+2} + \cdots + Y_n), \tag{2.86}$$

$$U_k := \frac{1}{\sqrt{n}}(X_1 + X_2 + \cdots + X_{k-1} + Y_{k+1} + Y_{k+2} + \cdots + Y_n). \tag{2.87}$$

Then $V_0 = S_n^Y/\sqrt{n}$ and $V_n = S_n^X/\sqrt{n}$. Since showing that $V_n \xrightarrow{d} V_0$ is equivalent to showing that $\mathbb{E}[f(V_n)] \to \mathbb{E}[f(V_0)]$ for all bounded smooth functions $f : \mathbb{R} \to \mathbb{R}$, it suffices to show

$$\mathbb{E}[f(S_n^X/\sqrt{n})] = \mathbb{E}[f(V_n)] \xrightarrow{d} \mathbb{E}[f(Z)] = \mathbb{E}[f(V_0)], \tag{2.88}$$

where f is a bounded smooth function and $Z \sim N(0,1)$. Now, we write $\mathbb{E}[f(V_n) - f(V_0)]$ as the telescoping sum $\sum_{k=1}^n \mathbb{E}[f(V_k) - f(V_{k-1})]$. Note that each term in this sum may be written as $\mathbb{E}[f(V_k) - f(V_{k-1})] = \mathbb{E}[f(V_k) - f(U_k)] - \mathbb{E}[f(V_{k-1}) - f(U_k)]$. To estimate the first of these terms, we use Taylor's theorem to write

$$f(V_k) - f(U_k) = f'(U_k)\frac{X_k}{\sqrt{n}} + f''(U_k)\frac{X_k^2}{2n} + f'''(W_k)\frac{X_k^3}{6n^{3/2}} \implies \mathbb{E}[f(V_k) - f(U_k)] = \mathbb{E}[f''(U_k)/2n] + (\cdots), \tag{2.89}$$

and similarly

$$f(V_{k-1}) - f(U_k) = f'(U_k)\frac{Y_k}{\sqrt{n}} + \frac{1}{2}f''(U_k)\frac{Y_k^2}{n} + \frac{1}{6}f'''(W_k')\frac{Y_k^3}{n^{3/2}} \implies \mathbb{E}[f(V_{k-1}) - f(U_k)] = \mathbb{E}[f''(U_k)/2n] + (\cdots), \tag{2.90}$$

which gives

$$|\mathbb{E}[f(V_k) - f(V_{k-1})]| \le |\mathbb{E}[f(V_k) - f(U_k)]| + |\mathbb{E}[f(V_{k-1}) - f(U_k)]| \le \frac{\|f'''\|_\infty}{6n^{3/2}}\mathbb{E}[|X_k|^3 + |Y_k|^3] \le \frac{C}{n^{3/2}}. \tag{2.91}$$

Therefore,

$$\mathbb{E}[f(V_n) - f(V_0)] = \sum_{k=1}^n \mathbb{E}[f(V_k) - f(V_{k-1})] \le \frac{C}{\sqrt{n}} \to 0, \tag{2.92}$$

as desired. ■

October 16th.

Theorem 2.23 (The *Linderberg-Feller central limit theorem*)**.** *Let $\{X_{ij}\}_{1 \le j \le i < \infty}$ be a triangular array of random variables such that:*

1. *For each $i \geq 1$, the random variables $X_{i1}, \ldots, X_{ii}$ are independent, with $\mathbb{E}[X_{ij}] = 0$.*
2. *$\sum_{j=1}^{i} \mathbb{E}[X_{ij}^2] \to \sigma^2 > 0$ as $i \to \infty$.*
3. *For all $\delta > 0$, $\sum_{j=1}^{i} \mathbb{E}[X_{ij}^2 \mathbf{1}_{|X_{ij}|>\delta}] \to 0$ as $i \to \infty$.*

Then, with $S_n = \sum_{j=1}^{n} X_{nj}$, we have $S_n \xrightarrow{d} N(0, \sigma^2)$ as $n \to \infty$.

We first show that this more general form of the central limit theorem implies the classical form. Let $X_1, X_2, \ldots$ be independent and identically distributed random variables with $\mathbb{E}[X_1] = 0$ and $\mathrm{Var}(X_1) = \sigma^2 > 0$. Then, for each $n \geq 1$, we can define a triangular array by setting $X_{nj} = X_j/\sqrt{n}$ for $1 \leq j \leq n$. Then, condition (1) is satisfied by independence, and condition (2) is satisfied since $\sum_{j=1}^{n} \mathbb{E}[X_{nj}^2] = \sum_{j=1}^{n} \mathrm{Var}(X_j/\sqrt{n}) = \sigma^2$. Finally, condition (3) is satisfied since

$$\sum_{j=1}^{n} \mathbb{E}[X_{nj}^2 \mathbf{1}_{|X_{nj}|>\delta}] = n\mathbb{E}[X_1^2 \mathbf{1}_{|X_1|>\delta\sqrt{n}}] \to 0, \tag{2.93}$$

as $n \to \infty$, by the dominated convergence theorem. This implies that $S_n = \sum_{j=1}^{n} X_{nj} = \sum_{j=1}^{n} X_j/\sqrt{n} \xrightarrow{d} N(0, \sigma^2)$ as $n \to \infty$, which is the classical form of the central limit theorem.

Example 2.24. Let $X_1, X_2, \ldots$ be independent random variables, with $X_n \sim \mathrm{Ber}(p_n)$ where $0 < p_n < 1$ for all $n \geq 1$. Let $\tau_n^2 = \sum_{j=1}^{n} \mathrm{Var}(X_j) = \sum_{j=1}^{n} p_j(1-p_j)$. Suppose $\tau_n^2 \to \infty$ as $n \to \infty$. Define $X_{nj} = (X_j - p_j)/\tau_n$ for $1 \leq j \leq n$. Then, we have $\mathbb{E}[X_{nj}] = 0$, satisfying condition (1). Also, $\sum_{j=1}^{n} \mathbb{E}[X_{nj}^2] = \sum_{j=1}^{n} \mathrm{Var}(X_j)/\tau_n^2 = 1$, satisfying condition (2). Finally, for any $\delta > 0$, we have

$$\sum_{j=1}^{n} \mathbb{E}[X_{nj}^2 \mathbf{1}_{|X_{nj}|>\delta}] = \sum_{j=1}^{n} \frac{\mathrm{Var}(X_j)}{\tau_n^2} \mathbf{1}_{|X_j - p_j|>\delta\tau_n} \leq \frac{1}{\tau_n^2} \sum_{j=1}^{n} \mathrm{Var}(X_j) = 1/\tau_n^2 \to 0, \tag{2.94}$$

as $n \to \infty$, satisfying condition (3). Therefore, by the Linderberg-Feller central limit theorem, we have

$$\frac{X_1 + X_2 + \cdots + X_n - \sum_{j=1}^{n} p_j}{\sqrt{\sum_{j=1}^{n} p_j(1-p_j)}} = \frac{S_n - \mathbb{E}[S_n]}{\sqrt{\mathrm{Var}(S_n)}} \xrightarrow{d} N(0,1), \tag{2.95}$$

as $n \to \infty$.

Example 2.25. Suppose $X_1, X_2, \ldots$ are independent and identically distributed random variables with $\mathbb{E}[X_1] = 0$ and $\mathrm{Var}(X_1) = 1$. Let $Y_n = X_1 + 2X_2 + \cdots + nX_n$, where $\mathrm{Var}(Y_n) = \sum_{j=1}^{n} j^2 \mathrm{Var}(X_j) = n(n+1)(2n+1)/6 \sim n^3/3$. We wish to show that

$$\frac{Y_n}{\sqrt{n^3/3}} \xrightarrow{d} N(0,1), \tag{2.96}$$

as $n \to \infty$. To see this, we define a triangular array by setting $X_{nj} = jX_j/\sqrt{n^3/3}$ for $1 \leq j \leq n$. Then, condition (1) is satisfied by independence, and condition (2) is satisfied since

$$\sum_{j=1}^{n} \mathbb{E}[X_{nj}^2] = \sum_{j=1}^{n} j^2 \mathrm{Var}(X_j)/(n^3/3) = \frac{3}{n^3} \sum_{j=1}^{n} j^2 \to 1, \tag{2.97}$$

as $n \to \infty$. Finally, for any $\delta > 0$, we have

$$\sum_{j=1}^{n} \mathbb{E}[X_{nj}^2 \mathbf{1}_{|X_{nj}|>\delta}] = \sum_{j=1}^{n} \frac{j^2}{n^3/3} \mathbb{E}[X_j^2 \mathbf{1}_{|X_j|>\delta\sqrt{n^3/3}/j}] \leq \frac{3}{n^3} \sum_{j=1}^{n} j^2 \mathbb{E}[X_j^2 \mathbf{1}_{|X_j|>\delta\sqrt{n}/\sqrt{3}}] = \mathbb{E}[X_1^2 \mathbf{1}_{|X_1|>\delta\sqrt{n}/\sqrt{3}}] \to 0, \tag{2.98}$$

as $n \to \infty$. Therefore, by the Linderberg-Feller central limit theorem, we have

$$\frac{Y_n}{\sqrt{n^3/3}} \xrightarrow{d} N(0,1), \tag{2.99}$$

as $n \to \infty$.

We now give a proof of the Linderberg-Feller central limit theorem.

Proof of the Linderberg-Feller central limit theorem. Let $S_n = X_{n1} + \cdots + X_{nn}$ be the sum of the n-th row. It suffices to show that, for all $t \in \mathbb{R}$, $\mathbb{E}[e^{itS_n}] \to \mathbb{E}[N(0,\sigma^2)]$ as $n \to \infty$. The Taylor expansion of e^{itx} about $x = 0$ gives

$$e^{itx} = 1 + itx + \frac{(it)^2}{2}x^2 + \frac{(it)^3}{3!}e^{itx^*}x^3 = 1 + itx + \frac{(it)^2}{2}e^{it\tilde{x}}x^2 \tag{2.100}$$

where $|\tilde{x}|, |x^*| \leq |x|$. We use the first expansion on $|x| < \delta$ and the second expansion on $|x| \geq \delta$; adding the two expansions gives

$$e^{itx} = 1 + itx - \frac{t^2}{2}x^2(1 - e^{it\tilde{x}})\mathbf{1}_{|x|\geq\delta} - \frac{it^3}{6}e^{itx^*}x^3\mathbf{1}_{|x|<\delta}. \tag{2.101}$$

Then, we have

$$\mathbb{E}[e^{itX_{nj}}] = 1 + 0 - \frac{t^2}{2}\mathbb{E}[X_{nj}^2] + R_{n,j} \tag{2.102}$$

where $R_{n,j}$ is small. We now show that if $z_j = 1 - \frac{t^2}{2}\mathbb{E}[X_{nj}^2]$ and $w_j = 1 - \frac{t^2}{2}\mathbb{E}[X_{nj}^2] + R_{n,j}$, then $\prod_{j=1}^n z_j \approx \prod_{j=1}^n w_j$, which happens when $\sum_{j=1}^n |R_{n,j}| \to 0$ as $n \to \infty$. ■

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